

FIXED-GEOMETRY BOTTLENECKS: DESIGNING PREDICTIVE ZERO-PENALTY ENDPOINTS

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Paper under double-blind review

ABSTRACT

Variational information bottlenecks penalize a representation’s KL divergence to a prior, and it is the prior that decides their high-regularization behavior: the fixed marginal prior of VIB places the zero-penalty point at an uninformative collapsed distribution, while the learned backward encoder of CEB leaves the zero-penalty set underdetermined, so the optimizer — not the objective — selects the final label geometry; moreover the object the bound certifies is the noisy training code, not the deterministic representation inference consumes. We take the zero-penalty set as the design object and propose the Fixed-Geometry Information Bottleneck (FGIB), which keeps the inference path deterministic and replaces the prior with a *frozen* label-conditional geometry — an orthonormal anchor frame for classification, fixed random-Fourier anchors for regression — reached through a side head that inference never reads. Because the reference cannot move, the compression term becomes a squared distance: at fixed classifier scale, on the class-symmetric aligned family, every β selects a distinct interior optimum whose fully compressed corner is the class-separated anchor configuration, with cross-entropy $\ln(1+(K-1)e^{-ca})$ rather than the chance-level $\ln K$ of a marginal prior. Two further properties follow by construction, and we prove them rather than leave them to observation. First, a deterministic main path *freezes the variance*: the KL’s gradient in the log-variance vanishes exactly at $\sigma^2 = \tau^2$, so the zero-initialized variance head never moves and the regularizer is *identically* the fixed-variance displacement term $\frac{1}{2\tau^2}\|\mu - av_y\|^2$ at every training step — releasing the variance is precisely what makes VIB and CEB race to the clamp, and FGIB eliminates the race by design. Second, in its canonical configuration the side head is a *fixed orthogonal* map ($\mu = h$ up to an isometry): the anchor matching then acts directly on the deployed representation with $\kappa(A^\top A) = 1$ and unit-covariance pull-back noise, closing the conditioning channel a trainable head leaves open. Across 9 datasets, 4 backbones and 5,100 training runs, under our grid and protocol FGIB attains the best mean rank among seven objectives — mean rank 1.86 on the seven classification settings (1.70 excluding AgeDB, best or tied-best on 6 of those 10), 1.60 on the five regression settings, and first also under the equal-budget control (mean rank 2.33 over all 12, a single globally fixed anchor scale) — and its β knob can be swept over a six-point grid spanning five orders of magnitude without destroying the model on classification, where VIB, CEB and DVCCA fall to the trivial predictor. A 400-epoch control finds CEB’s variance drifting exactly in the race’s predicted direction while FGIB’s stays pinned at τ^2 ; a subject-disjoint AgeDB split preserves the regression ordering; and the fixed-orthogonal head carries a small measured price across settings, which we quantify rather than hide. The KL’s role is an exact variational upper-bound interpretation for the auxiliary noisy code, not a certificate of the deployed representation’s compression. The guarantee weakens on regression, where the continuous-anchor construction is an exploratory transfer rather than a proof: we keep an explicit ledger of what is proved and what is open.

1 INTRODUCTION

An information bottleneck asks a representation Z of an input X to be maximally informative about a target Y and minimally informative about everything else (Tishby et al., 1999). The deep variational instantiation (Alemi et al., 2017) replaces the intractable $I(X; Z)$ with $\mathbb{E} \text{KL}(q(z|x) \| m(z))$ and minimizes $\text{CE} + \beta \text{KL}$; the Conditional Entropy Bottleneck (Fischer, 2020) sharpens the target to the *residual* $I(X; Z|Y)$ with a learned backward encoder $b(z|y)$. The programme has an appealing shape: a single knob β should trace a trade-off curve, and the value of the KL should serve as a meter reporting how much unnecessary information the model still carries.

Two properties of that construction are usually left implicit, and they interact badly. First, *where the zero-penalty set lives*: the KL vanishes exactly on the prior, so the prior chooses the set of representations the regularizer never penalizes — a marginal prior places it at collapse, a learned conditional prior leaves it an underdetermined family, and the optimizer, not the objective, selects the final label geometry. Second, *the object being regularized is not the object being deployed*: the classifier is trained on a sampled code $z \sim q(z|x)$ and evaluated on the posterior mean $\mu(x)$ (Fischer, 2020, App. A.1), so the bound constrains a noisy variable that inference never sees.

This paper starts from a diagnosis of what those two properties cost, and asks what changes if the prior’s geometry is fixed by design rather than left to move or to vanish.

The diagnosis. We isolate three failure channels of the conditional bottleneck and one structural mismatch; all are stated and proved in Appendix A, and Section 2 summarizes them. The first two are properties of the objective, not of who is trainable: they hold at the level of the information plane before any optimizer runs. (P1) *The knob is dead*: in the deterministic scenario every $\gamma > 0$ of CEB has the *same* exact minimizer, the copy corner (Proposition 1), so the knob cannot be reading off a position on the IB curve. (P2) *The certificate is blind on the necessity axis*: the residual certificate ReX is identically 0 on the collapse-to-copy manifold, and the objective trades one nat of label information for $1/\beta$ nats of certified compression (Proposition 2). (P3) *The variances race — in the gradient, and slowly in our longest run*: with trainable variances the KL is stationary at $\sigma^2 = \tau^2$ while the cross-entropy pushes σ^2 down without opposition once the means match, so the predicted equilibrium sits at the clamp (Proposition 3); the prediction is directly testable and we ran the test — on the full validation set of every epoch, out to a 400-epoch control. Under the 100-epoch protocol the race is invisible; at 400 epochs CEB’s variance drifts monotonically in exactly the predicted direction (-2.3 nats at $\beta = 1$), still far from the clamp — real but slow (Section 4.5). For FGIB the question is not empirical at all: a deterministic main path makes the KL’s log-variance gradient vanish exactly at $\sigma^2 = \tau^2$, pinning the variance there for the whole run, so the regularizer is *identically* the fixed-variance displacement term $\frac{1}{2\tau^2} \|\mu - av_y\|^2$ — the race is eliminated by design (Proposition 7), and its only re-entry point, a trainable prior variance, is the one FGIB-family variant that collapses (Corollary 1). (P4) *The certified object is the wrong one*: no finite certificate bounds the residual of a deployed deterministic representation — CEB’s or ours (Proposition 4).

The method. FGIB (Section 3) makes three changes that follow from the diagnosis: (i) the inference path is deterministic — the classifier consumes the encoder output h at training *and* at evaluation; (ii) the stochastic bottleneck moves to a *side path* ($\mu(x) = Ah(x)$ plus a log-variance head) that no inference path ever reads, keeping the variational bound finite without putting noise in front of the classifier; (iii) the label-conditional prior is *fixed by design*, $r(z|y) = \mathcal{N}(av_y, \tau^2 I)$ with an orthonormal anchor frame, so the zero-penalty set is the prescribed class-separated configuration rather than collapse or an underdetermined family (regression: fixed random-Fourier anchors on a sphere of radius a).

Freezing the reference is what buys the theory. Against a fixed μ_p , the KL’s displacement term $\frac{1}{2\tau^2} \|\mu - \mu_p\|^2$ is *quadratic* in the alignment scalar — the convexification squared-IB (Kolchinsky et al., 2019a) installs by hand is built in — and at fixed classifier scale every β has a unique interior optimum on a strictly concave trade-off curve (Propositions 8 and 9). Equally important, the KL now vanishes at the class-separated configuration $\mu_y = av_y$, with cross-entropy $\ln(1 + (K-1)e^{-ca}) \rightarrow 0$ as a grows, where a marginal $\mathcal{N}(0, I)$ prior places $\text{KL} = 0$ at $\mu \equiv 0$ with cross-entropy $\ln K$ — chance (Remark 13). *The degenerate corner is excluded by geometry rather than by the optimizer’s good behavior.*

The evidence. We evaluate on 9 datasets over MLP, CNN, GCN and LSTM backbones (12 task×backbone settings) against six objectives: the deterministic backbone, VIB (Alemi et al., 2017), squared-IB (Kolchinsky et al., 2019a), nonlinear IB (Kolchinsky et al., 2019b), CEB (Fischer, 2020) and supervised β -DVCCA (Abdelaleem et al., 2025). The sweep is 1,020 configurations $\times$ 5 seeds = 5,100 training runs, with every hyperparameter selected on validation. Three findings, in Section 4:

- FGIB attains the best mean rank (1.75 of 7) and is best or tied-best on 7 of 12 settings; the next best entry is the deterministic backbone at 3.42, and all five prior bottlenecks fall between 3.75 and 5.08 (Table 1). The rank decomposes as 1.86 on the seven classification settings, 1.60 on the five regression settings, and 1.70 (best or tied-best on 6 of 10) excluding the two AgeDB rows, whose image-level split leaks identity across the boundary (Appendix G). Outside the two settings with a weak backbone, no bottleneck — FGIB included — improves accuracy by more than about a point: the headline number is a rank, not a margin.
- Sweeping β over a six-point grid spanning five orders of magnitude leaves FGIB within 0.52 accuracy points of its backbone on *every* classification setting, at a single globally fixed anchor scale, while VIB, CEB and DVCCA fall to the trivial predictor on all seven (Table 2). NIB is equally stable but largely inert: its median gain at its *best* β is -0.27 points, so at the median its knob buys nothing. FGIB is the only objective in the study with both a non-destructive and a positive-median-gain knob — the former robust, the latter task-set-sensitive (Section 4).
- The mechanism is the one the theory names. At maximum compression pressure the anchor scale a controls whether the model survives: at $a = 1$ — where the anchors have nearly collapsed onto the single origin of a VIB prior (Remark 22) — FGIB falls to chance on the 100-class task (1.0%) and on Cora (26.6%), and climbs back to 83.0% and 87.2% as a grows (Table 3). The 100-class task needs the largest anchor, as the $(K-1)$ factor in the corner value predicts.

What we do not claim. Three of the failure channels are addressed by FGIB’s construction; the object mismatch is not, and one of the closures is conditional. In its *canonical* configuration the side head is a fixed orthogonal map, anchoring the deployed representation directly with $\kappa(A^\top A) = 1$ (Proposition 6); a trainable side head is an *optional* variant (Remark 7; its conditioning cost is measured in Section 4.5, and its price for the closure is quantified in Table 8) — only the fixed-orthogonal configuration is covered by the closure claim. The one-to-one sweep requires a bounded classifier scale that the present implementation does not enforce (Remark 12; enforced variants are tested in Section 4.5); no finite certificate bounds the deployed deterministic residual, ours included (Proposition 4); and the sweep is a statement about a KL–CE surrogate curve on a class-symmetric family, not a recovered IB curve (Remark 11). The robustness of the regularizer is carried by the anchor geometry rather than by the KL machinery or the frozenness of the anchor means (Section 4.5); what the KL adds is the residual-information reading for the auxiliary code and the sweep theory, not a measured accuracy margin, and its realized objective is exactly a fixed-anchor center loss — an identity, not an approximation (Proposition 7). Appendix A keeps a complete ledger of what is proved and what is open.

2 PRELIMINARIES: THREE FAILURE CHANNELS AND AN OBJECT MISMATCH

X is the input, Y the label with K classes, and an encoder induces the Markov chain $Y - X - Z$; throughout we use the loss convention $\text{CE} + \beta \text{KL}$ ($\beta = 1/\gamma$, $\beta_{\text{IB}} = 1 + 1/\beta$). The variational objects $R = \mathbb{E}[\text{KL}(q(z|x)||m(z))]$ $\geq I(X; Z)$ (VIB’s rate), $\text{ReX} = \mathbb{E}[\text{KL}(q(z|x)||b_\theta(z|y))]$ $\geq I(X; Z|Y)$ (CEB’s residual, trainable b_θ), and ReX_F with a *fixed* anchor r are instances of the gap identity $\mathbb{E}[\text{KL}(q(z|x)||r(z|y))] = I(X; Z|Y) + \mathbb{E}_y[\text{KL}(q(z|y)||r(z|y))]$ (Lemma 4): a bound plus a slack, and everything below turns on *who is allowed to move that slack to zero* — CEB’s backward encoder can do it with the encoder held fixed, so its bound can be tight while the representation is entirely uncompressed. Four channels follow (proved in Appendix A, stated in detail in Appendix B): (P1) *The knob is dead*: in the deterministic scenario every $\gamma > 0$ of CEB has the same exact minimizer, the copy corner, so the Lagrangian never sweeps the IB curve (Proposition 1). (P2) *The certificate is blind on the necessity axis*: ReX is identically 0 on the collapse-to-copy manifold T_α from

total collapse to full copy, and the objective trades one nat of label information for $1/\beta$ nats of certified compression (Proposition 2). (P3) *The variances race*: the KL is stationary in σ^2 at $\sigma^2 = \tau^2$ while the cross-entropy strictly prefers $\sigma^2 \rightarrow 0$, so once the means match the race runs to the clamp unopposed, degenerating the certificate into a mean-matching force of weight e^{10} (Proposition 3); the prediction is a direction-of-force argument. Measured on the full validation set, the race is invisible under the 100-epoch protocol but its direction shows up clearly by 400 epochs (Section 4.5, Remark 4). (P4) *The certified object is not the deployed object*: $I(X; Z|Y) \leq I(X; \mu(X)|Y) = +\infty$ for any continuous non-class-constant μ , so no finite certificate bounds the deployed deterministic residual — CEB’s or FGIB’s (Proposition 4).

3 METHOD: THE FIXED-GEOMETRY INFORMATION BOTTLENECK

3.1 MODEL

A shared encoder computes the deterministic representation $h = f_{\text{enc}}(X)$, and the task head consumes h directly at training and at evaluation time: logits = $c(h)$. In parallel, a *side path* maps h to a diagonal Gaussian posterior

$$q(z|x) = \mathcal{N}(\mu(x), \Sigma(x)), \quad \mu(x) = Ah(x), \quad \Sigma(x) = \text{diag} \exp(\ell(h(x))), \quad (1)$$

with a linear mean head A and a log-variance head ℓ , neither evaluated at inference. The label-conditional prior is a set of frozen buffers,

$$r(z|y) = \mathcal{N}(av_y, \tau^2 I), \quad V = [v_1, \dots, v_K] \text{ orthonormal}, \quad V^\top V = I_K, \quad (2)$$

where V is the orthonormal factor of the QR decomposition of a random matrix, $a > 0$ the *anchor scale* and τ^2 the *anchor variance* (default 1). The training objective is

$$L_{\text{FGIB}} = \mathbb{E}[\text{CE}(c(h), y)] + \beta \mathbb{E}[\text{KL}(q(z|x) \| r(z|y))]. \quad (3)$$

Gradients from the second term reach the main path only through the shared encoder. The log-variance head is zero-initialized, so training starts at $\sigma^2 = \tau^2$ and $\text{KL} \approx a^2/(2\tau^2)$ — and it does not merely *start* there: Result 2 below proves the variance stays pinned at τ^2 for the entire run, so the realized regularizer is exactly $\frac{1}{2\tau^2} \|\mu - \mu_p\|^2$ at every step. The canonical configuration fixes the side head to an *orthogonal* map — with $\dim z = \dim h$ one may take $A = I$ without loss (Result 3) — so the anchor matching acts directly on the deployed representation; the trainable head of Eq. (1) is then an optional ablation, and we call the canonical form *FGIB-H* ($A = I$). For regression the per-class table is replaced by fixed random-Fourier anchors (Rahimi & Recht, 2008), $\mu_p(y) = a\sqrt{2/\dim z} \cos(2\pi\omega y + b)$ with $\omega \sim \mathcal{N}(0, 4)$ and $b \sim U[0, 2\pi)$ frozen at initialization, so that $\|\mu_p(y)\| \approx a$; setting $a = 0$ recovers a shared $\mathcal{N}(0, \tau^2 I)$ prior, i.e. a VIB-style marginal (Remark 22). Three design decisions distinguish this from CEB, each targeting one of P1–P4: the classifier consumes h rather than z (P3, P4); the two loss terms act on disjoint parameter sets (P3); the reference is a buffer rather than a network (P1, P2).

Theory–implementation layering. The analysis idealizes the side head as a linear invertible map with $\dim z = \dim h$; the implementation uses an `nn.Linear with bias` and no orthogonality, invertibility or conditioning constraint — and that freedom is precisely what lets the trainable head absorb the anchor fit without moving h (Section 3.3(a), measured in Section 4.5). The canonical FGIB-H configuration removes the gap rather than flagging it: $A = I$ is orthogonal and invertible by construction, so the idealizations below — invertibility, conditioning, the pull-back reference of Result 1 — hold exactly in the implementation; for the trainable head they hold as approximations, and Section 3.3(a) states how the gap can widen when A is freed. A second layering gap concerns the variational family: the exact KL invariance of Result 1 holds for the underlying Gaussians, but the diagonal family is not closed under A^{-1} , so within it the identity is exact distributionally and approximate as a variational identity (Remark 8).

3.2 WHAT THE FROZEN REFERENCE BUYS

Result 1 (The side path is a CEB bottleneck on the deployed representation). *Let $\dim z = \dim h$ and let A be invertible. Pulling back by A^{-1} maps the posterior to $q'(\tilde{z}|x) =$*

$\mathcal{N}(h(x), A^{-1}\Sigma A^{-\top})$ — centered exactly at the representation inference uses — and the anchors to $\mathcal{N}(aA^{-1}v_y, \tau^2(A^\top A)^{-1})$, which are orthonormal at scale a in the induced metric $A^\top A$. Since the Gaussian KL is invariant under a common invertible linear map (Lemma 1), the regularizer of Eq. (3) is pointwise identical to a CEB bottleneck — with the backward encoder frozen at the current side head — applied to $\tilde{z} = h + \tilde{\varepsilon}$ (Proposition 5). At the variance optimum $\Sigma = \tau^2 I$ the perturbation is $\tau(A^\top A)^{-1/2}\varepsilon$: known, side-path-only noise whose covariance is a model parameter (Proposition 11).

This licenses calling the construction a bottleneck at all: the regularizer carries a residual-information bound read in coordinates centered on the deployed representation — an interpretation a bare prototype loss lacks. The identity holds pointwise at each value of A ; because a trainable A moves, the pulled-back reference moves with it — the one channel the trainable-head configuration leaves open, closed by the canonical configuration (Result 3).

Result 2 (A deterministic main path freezes the variance: the realized KL is a fixed-anchor center loss). *With the classifier consuming h (never z), a zero-initialized log-variance head, a frozen prior log-variance $\ell_p = \ln \tau^2 = 0$, and a first-order optimizer without weight decay, the posterior log-variance is identically $\ell_q \equiv \ell_p$ at every training step, and the regularizer is exactly $\text{KL}(q \| r) = \frac{1}{2\tau^2} \|\mu - \mu_p\|^2$ — the KL of Result 4 evaluated permanently at the variance optimum, not merely in the limit. More generally, the per-coordinate gap $u = \ell_q - \ell_p$ obeys $\partial_u \text{KL} = \frac{1}{2}(e^u - 1)$, which carries the sign of u , so $u = 0$ is the unique attractor of any first-order descent: releasing the variance is exactly what makes VIB and CEB race to the clamp (Channel P3), and the deterministic main path eliminates the race by design (Proposition 7). The same argument pins the posterior variance in every FGIB-family ablation, so FGIB, TAFGIB and CentFGIB are exactly the same loss up to the trainability of the anchor means — the ablation table’s number-coincidence is an identity — and the one variant that escapes the family, DCEB’s trainable prior variance, is the one that collapses, at $\beta = 10$: the race re-enters through the prior (Corollary 1).*

Result 3 (A fixed orthogonal side head anchors the deployed representation). *Let the side head be a fixed orthogonal map ($\dim z = \dim h$). Then $\|Ah - av_y\| = \|h - aA^\top v_y\|$, the rotated frame $\{A^\top v_y\}$ is orthonormal, and the regularizer is a fixed-anchor center loss on the deployed representation h itself. The induced metric of Result 1 is $A^\top A = I$ with $\kappa = 1$ at every step: the pulled-back reference is orthonormal at scale a , the perturbation is exactly $\tau \varepsilon$ with unit covariance, and the conditioning counterexample of Proposition 12(iii) is excluded by construction. For Haar-distributed random orthogonal A the rotated frame is identically distributed to $\{v_y\}$ across seeds, so $A = I$ is a representative canonical choice (Proposition 6). We call the canonical form FGIB-H and measure its small price across settings (Section 4.5, Table 8).*

Result 4 (Every β selects a distinct interior optimum). *Consider the class-symmetric aligned family — the encoder maps class y to $h_y = s v_y$, the classifier is held at $w_j = c v_j$ with fixed scale c . Then*

$$\text{KL}(s, \sigma^2) = \frac{\dim z}{2} \left(\frac{\sigma^2}{\tau^2} - 1 - \ln \frac{\sigma^2}{\tau^2} \right) + \frac{1}{2\tau^2} (s - a)^2, \quad \text{CE}(s) = \ln(1 + (K - 1)e^{-cs}), \quad (4)$$

and the variance optimum is $\sigma^{2} = \tau^2$ for every β , because the prediction term does not see σ^2 (Lemma 3). The Lagrangian $L_\beta(s) = \text{CE}(s) + \frac{\beta}{2\tau^2} (s - a)^2$ is then strictly convex with a unique minimizer $s^*(\beta) > a$ given by $\frac{\beta}{\tau^2} (s - a) = c \varphi(cs)$, $\varphi(u) = (K - 1)e^{-u} / (1 + (K - 1)e^{-u})$; the map $\beta \mapsto s^*(\beta)$ is continuous and strictly decreasing, with $s^* \rightarrow a$ as $\beta \rightarrow \infty$ and $s^* \rightarrow \infty$ as $\beta \rightarrow 0$, so $\beta \mapsto (\text{KL}(s^*), \text{CE}(s^*))$ is one-to-one onto the entire curve, which is strictly concave (Propositions 8, 9).*

The mechanism is the whole point of freezing the reference. Deterministic-IB on hard clusterings has compression and prediction as the *same* scalar, so the objective is monotone in it; standard IB in the deterministic scenario has a piecewise-linear curve with a single supporting slope. Both are repaired in Kolchinsky et al. (2019a) by *squaring* the compression term by hand. FGIB needs no surgery: $\text{KL} = \frac{1}{2\tau^2} \|\mu - \mu_p\|^2 + (\text{variance terms})$ is quadratic in the displacement *whenever μ_p does not move*. Entropy-based compression is self-referential — it depends only on the representation’s own distribution; a fixed anchor supplies an absolute frame and turns compression into a squared distance. The convexification is inherited, not installed.

Result 5 (A persistent force and a non-collapsed compressed endpoint). *At a fixed encoder, CEB’s bound is tighter in value: $\text{ReX}(\theta^*) \leq \text{ReX}_F$, the excess being the anchor gap*

$\mathbb{E}_y[\text{KL}(q(z|y) \|\mathcal{N}(av_y, \tau^2 I))] - \mathbb{E}_y[\text{KL}(q(z|y) \|\mathcal{N}(m_y, S_y))] \geq 0$ (Proposition 10). That excess is the price of the fixed anchor, and it is paid for a reason: CEB’s gap is a function of the backward encoder alone, so it can be driven to zero with the representation untouched, whereas FGIB’s gap can shrink only if the posterior moves toward the anchors — the force is persistent (Remark 17). Moreover, the $\text{KL} = 0$ point is $\mu_y = av_y$, a class-separated configuration with $\text{CE} = \ln(1 + (K-1)e^{-ca})$, which is bounded and $\rightarrow 0$ as a grows; a marginal $\mathcal{N}(0, I)$ prior instead places $\text{KL} = 0$ at $\mu \equiv 0$ with $\text{CE} = \ln K$, i.e. chance, and squaring the KL (SVIB) leaves that corner in place since KL^2 is stationary at $\text{KL} = 0$ (Remark 13). The trivial collapse pays a fixed positive price $\frac{1}{2\tau^2}a^2$ per class.

Result 5 is the prediction the experiments test most directly: it says that pushing β up should not destroy an FGIB model, and that the anchor scale a is the dial controlling how benign the fully compressed corner is. Both are visible in Tables 2 and 3.

3.3 WHAT THE FROZEN REFERENCE DOES NOT BUY

We state the open items here rather than in a closing paragraph, because two of them bear on how the experiments should be read. (a) *A trainable side head is a second trainable party in the KL* — structurally CEB’s backward encoder — and it can absorb the anchor fit without moving h ; the canonical fixed-orthogonal head closes this channel by construction, and the measurement of Section 4.5 shows the trainable variant’s $\kappa(A^\top A)$ degrades to $10^{5.8}$ – $10^{6.8}$ while accuracy is unaffected, so the freedom buys nothing (Appendix C). (b) *The sweep needs a bounded classifier scale c* ; the implementation leaves it free, so Result 4 is conditional as implemented, and enforced-scale variants are tested on MNIST (Section 4.5). (c) *The certificate still does not bound the deployed residual* (P4 applies unchanged). (d) *The compressed endpoint is designed, not certified*: the anchor corner is a label code, not semantic compression (Remark 13). (e) *The sweep is over a surrogate*: the $(\text{KL}_{\text{anchor}}, \text{CE})$ plane on the aligned family, not the information plane (Remark 11). The full statement of each item is in Appendix C.

4 EXPERIMENTS

4.1 SETUP

Tasks and backbones. We use 9 datasets over 4 backbones, giving 12 task×backbone settings: MNIST (LeCun et al., 1998) (MLP/CNN), ImageNet-100 (Deng et al., 2009) frozen ResNet-50 (He et al., 2016) layer-3 features pooled to $1024 \times 4 \times 4$ (MLP/CNN), Cora (Sen et al., 2008) transductive node classification (GCN (Kipf & Welling, 2017)), IMDb (Maas et al., 2011) and AG News (Zhang et al., 2015) (frozen BERT (Devlin et al., 2019) token features, PCA-50) with LSTM (Hochreiter & Schmidhuber, 1997) encoders; and the regressions California Housing (Pace & Barry, 1997) (MLP), STS-B (Cer et al., 2017; Wang et al., 2019) (GloVe (Pennington et al., 2014) LSTM), ZINC-12k (Dwivedi et al., 2023) graph-level regression (GCN), and AgeDB (Moschoglou et al., 2017) face-age regression at 64×64 RGB (MLP/CNN). Splits are 60/20/20 (seed 42) except ZINC (official 10,000/1,000/1,000); several splits are custom, so every comparison in this paper is internal to the same split. AgeDB’s image-level split leaks identity across the boundary (562 of 567 subjects); a subject-disjoint control, run here, preserves the objective ordering (Table 9), so the leak inflates magnitudes but does not create the pattern. Appendix G gives the preprocessing.

Objectives. Each backbone carries the same seven heads: the deterministic backbone, VIB (Alemi et al., 2017), squared-IB (Kolchinsky et al., 2019a, SVIB, penalty βKL^2), nonlinear IB (Kolchinsky et al., 2019b, NIB, non-parametric pairwise bound, learned noise variance), CEB (Fischer, 2020, diagonal-Gaussian backward encoder), supervised β -DVCCA (Abdelaleem et al., 2025, VIB plus an input-reconstruction decoder), and FGIB. For regression, CEB and FGIB condition their priors on the min-max-scaled continuous target. Every variant shares the encoder, the bottleneck dimension ($\dim z = 256$) and the optimizer. The objectives’ compression terms are not dimensionally identical (NIB’s bound is in nats; DVCCA’s β also weights reconstruction), so β is treated as a per-objective knob swept independently and selected on validation, never as a shared axis of equal compression.

Table 1: Test performance of the validation-selected configuration of each objective, averaged over 5 runs (seeds 0–4). Accuracy (%) for classification, $R^2 \times 100$ for regression; higher is better. *Det.* is the deterministic backbone trained with the task loss alone. For every bottleneck β is chosen per cell by mean best-epoch validation score — and for FGIB the anchor scale a as well — never by test score, so FGIB is selected over a 6×9 grid and the others over a 6-point grid; exact ties are broken by a second validation quantity and then by fixed grid order. Best per row in bold (ties within 0.01 points both bolded); per-run standard deviations and the selected hyperparameters are in Table 4, whose last column reports the equal-budget FGIB control.

Task	Backbone	Det.	VIB	SVIB	NIB	CEB	DVCCA	FGIB
<i>Classification (Acc %)</i>								
MNIST	MLP	98.33	98.28	98.10	98.26	98.28	98.24	98.49
MNIST	CNN	99.13	99.28	99.19	99.19	99.25	99.28	99.27
ImageNet-100	MLP	83.12	82.65	83.02	82.34	82.56	82.56	83.96
ImageNet-100	CNN	83.40	82.94	82.61	82.72	82.33	83.09	83.55
Cora	GCN	87.68	86.83	77.56	86.35	86.27	86.83	87.68
IMDb	LSTM	84.85	85.43	85.99	86.11	85.55	85.69	86.04
AG News	LSTM	92.94	92.61	92.70	92.91	92.66	92.78	92.71
<i>Regression ($R^2 \times 100$)</i>								
Cal. Housing	MLP	76.18	79.01	78.97	78.62	78.94	78.78	79.85
STS-B	LSTM	28.99	27.19	26.97	24.17	26.68	27.05	29.35
ZINC-12k	GCN	63.48	63.39	63.25	61.78	62.57	62.94	63.42
AgeDB	MLP	21.80	31.73	33.85	27.10	33.17	31.80	32.71
AgeDB	CNN	39.14	33.42	33.23	33.37	38.82	32.05	39.56
Mean rank (of 7)		3.42	3.75	4.58	5.08	4.75	4.17	1.75
Best or tied-best (of 12)		3	1	1	1	0	1	7

Protocol. Adam (Kingma & Ba, 2015) at 10^{-3} (10^{-2} for Cora), batch size 512 (128 for ZINC), at most 100 epochs, early stopping with patience 10 on validation macro-AUC (classification) or validation R^2 (regression); each configuration is run 5 times with seeds 0..4 and we report the mean over runs. The grid is $\beta \in \{10^{-4}, \dots, 10\}$ for every bottleneck plus $a \in \{1, \dots, 16\}$ for FGIB: 1,020 configurations, 5,100 runs. **Hyperparameters are selected on validation, never on test:** each cell reports the test score of its best mean-validation configuration, with exact ties broken by a second validation quantity and then fixed grid order, so no selection step consults the test set. FGIB is thus selected over a 6×9 grid where the others use 6 points — a search-budget advantage we return to with an equal-budget control. Per-run standard deviations and selected hyperparameters are in Appendix D; the ablations of Section 4.5 use the identical protocol on MNIST/MLP.

4.2 ACCURACY: FGIB RANKS FIRST, BUT NO BOTTLENECK WINS BY MUCH

Table 1 reports the validation-selected configuration of each objective. FGIB attains the best mean rank, 1.75 of 7, and is best or tied-best on 7 of the 12 settings; the runner-up is the deterministic backbone at 3.42, and the five prior bottlenecks occupy 3.75 (VIB) to 5.08 (NIB). The wins span every modality and backbone, and where FGIB does not win it is close (within 0.07 points on MNIST/CNN, IMDb and ZINC); it is clearly beaten only on AG News (92.71 vs. 92.94) and AgeDB/MLP (32.71 vs. 33.85).

Three qualifications are essential. First, *the margins are small*. Excluding the two settings where every bottleneck helps a weak backbone, no objective *improves* on its backbone by more than about one point, and on 3 of 12 settings the deterministic backbone is still best or tied-best; losses can be large even at the validation-selected optimum (SVIB gives up 10.1 points on Cora). Reading Table 1 as evidence that bottlenecks substantially improve in-distribution accuracy is reading it wrong; it shows that FGIB is the only one that is reliably not *worse*. Second, *FGIB’s selection grid is $9 \times$ larger*. The equal-budget control — the last column of Table 4, the same 6-point β grid at a fixed $a = 16$ (post-hoc) — has mean rank 2.33 and best-or-tied 6 of 12, still ahead of the backbone (3.42) and every prior bottleneck (3.75–5.08): part of the rank advantage is search budget; the ordering of methods is not (Table 6 shows the fixed-anchor rank ranges from 2.00 at $a = 14$ to 3.17 at $a = 1$). Third, *the rank gaps are backed by paired statistics, and we report their sensitivity*.

Table 2: Robustness of the compression knob. For each objective we sweep $\beta \in \{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1, 10\}$ and report the *worst* test score any β produces, as a deficit (in points) against the deterministic backbone of the same row; negative means the objective never falls below its backbone. Units as in Table 1. FGIB is shown twice: at the anchor scale selected on validation in Table 1, and at a single globally fixed anchor scale $a = 16$ (a post-hoc sensitivity control, not selected a priori). Lower is better; best per row in bold. The summary rows take medians over the 12 settings and count settings where the worst case falls more than 5 points below the backbone. The “best β ” gain row is the maximum over the grid of the *test* score — an exploratory upper bound, not a model-selection result; the validation-selected gain row is the selection result of Table 1 expressed relative to the backbone.

Task	Backbone	VIB	SVIB	NIB	CEB	DVCCA	FGIB	
							val-sel. a	$a=16$
<i>Classification (Acc %)</i>								
MNIST	MLP	86.98	79.15	0.18	87.69	86.98	0.02	0.02
MNIST	CNN	87.78	66.25	0.00	87.78	87.78	-0.02	-0.02
ImageNet-100	MLP	82.09	82.09	0.86	82.09	82.09	0.40	0.17
ImageNet-100	CNN	82.32	23.34	0.85	82.37	82.37	0.18	0.41
Cora	GCN	57.46	57.42	1.33	64.80	66.50	61.11	0.52
IMDb	LSTM	34.85	0.36	-0.92	34.85	34.76	-0.08	-0.45
AG News	LSTM	67.63	0.41	0.22	67.33	67.94	0.23	0.29
<i>Regression ($R^2 \times 100$)</i>								
Cal. Housing	MLP	76.20	76.15	-2.12	76.21	76.20	-1.99	-1.78
STS-B	LSTM	185.34	119.84	6.86	30.13	185.91	13.21	13.21
ZINC-12k	GCN	63.48	63.39	2.04	58.59	334.43	3.27	4.54
AgeDB	MLP	21.88	21.91	-4.72	21.93	21.88	21.87	24.04
AgeDB	CNN	60.18	70.86	5.78	39.23	59.71	15.92	15.57
Median deficit		71.91	64.82	0.53	66.06	79.14	0.31	0.35
Settings destroyed (of 12)		12	10	2	12	12	4	3
Median gain at best β (test-selected upper bound)		-0.15	-0.14	-0.27	-0.22	-0.15	0.39	0.27
Median gain at validation-selected β		-0.21	-0.23	-0.37	-0.30	-0.24	0.26	0.20

Table 5 reports per-seed paired differences against the backbone with bootstrap 95% CIs (over the 5 paired seeds; the intervals describe training randomness, not split uncertainty). A Wilcoxon test over the 12 setting-level paired means finds FGIB the only objective whose gain over its backbone is significantly positive ($p = 0.014$; $p = 0.12$ at the equal-budget control), and FGIB beats each prior bottleneck pairwise ($p \leq 0.016$, all significant under Holm). The gain result is task-set-sensitive, and we say so plainly: excluding the two AgeDB rows the mean paired gain falls from +1.46 to +0.62 points and the Wilcoxon p -value from 0.014 to 0.055, while the mean rank (1.70, best-or-tied on 6 of 10 settings) and the stability results are unchanged; a subject-disjoint AgeDB split preserves the objective ordering (Table 9), so the leak inflates magnitudes without creating the pattern. We therefore take the *rank* and the *non-destructiveness of the knob* as the primary statistical claims, treat the gain-over-backbone as marginal and sensitive, and exclude AgeDB from the core statistical statements of this section.

4.3 THE KNOB: FGIB IS THE ONLY OBJECTIVE WHOSE β IS BOTH SAFE AND USEFUL

Result 5 predicts what happens at the *other* end of the sweep: a marginal-prior bottleneck has its $KL = 0$ point at $\mu \equiv 0$, so raising β far enough must drive it to the trivial predictor; FGIB’s $KL = 0$ point is class-separated. Table 2 measures exactly this: for each objective we take the *worst* test score produced by any β in the grid, and report the deficit against the deterministic backbone of the same row.

The separation is stark. VIB, CEB and DVCCA are driven below the trivial predictor on all 7 classification settings (median worst-case deficits 71.9, 66.1, 79.1 points); SVIB is only marginally better (64.8), as Remark 13 predicts: squaring the KL does not move the corner. FGIB’s median worst-case deficit is 0.31 points at the validation-selected anchor and 0.35 at fixed $a = 16$; on

classification the worst case over the six-point, five-order-of-magnitude β grid is within 0.52 points of the backbone on *every* one of the seven settings (per-setting values in Table 2) — against deficits of 34–88 points for VIB, CEB and DVCCA on the same rows.

NIB is the interesting control. It is as stable as FGIB (median deficit 0.53), but for a different reason and at a different price: its non-parametric bound saturates at $\ln N$ with exponentially vanishing gradient once the noise is small relative to the cluster spread (Kolchinsky et al., 2019b, §IV A), so it is stable largely because the knob barely acts — on MNIST/MLP its accuracy moves from 98.15 to 98.27 across the entire sweep. Table 2 reports two gain summaries, and the distinction matters. “Median gain at best β ” takes the maximum over the grid of the *test* score — an exploratory upper bound, not a model-selection result — and on it NIB scores -0.27 points: even with test-selected optimism the knob is not worth turning, and VIB, SVIB, CEB, DVCCA (-0.15 , -0.14 , -0.22 , -0.15) are likewise negative; only FGIB is positive ($+0.39$; $+0.27$ at $a = 16$). The protocol-clean row is the median gain at the *validation-selected* configuration of Table 1: FGIB $+0.26$ ($+0.20$ at $a = 16$), the five prior bottlenecks -0.21 to -0.37 — the conclusion survives the protocol correction. *Among the six bottlenecks, FGIB is the only one whose compression knob is both non-destructive and, at the median, worth turning.*

The exceptions are informative and we do not hide them. FGIB is destroyed — worst case worse than the backbone by more than 5 points — in 3 of 12 settings at $a = 16$, and all three are regression: STS-B (13.2 points of R^2), AgeDB (15.6 CNN, 24.0 MLP), with ZINC just below threshold at 4.5. We therefore report the regression results as *exploratory* and keep the core claims on classification. The theory anticipates this asymmetry: the random-Fourier anchor construction for continuous targets is a transfer argument, not a proof, and it inherits both the fixed-scale and the conditioning caveats (Remark 16). Whatever protects the compressed corner in the orthonormal per-class case does not transfer intact to the RFF case.

4.4 THE ANCHOR SCALE IS THE DIAL THE THEORY SAYS IT IS

Result 5 says *why* FGIB is robust, and therefore when robustness should fail: the compressed corner costs $CE = \ln(1 + (K-1)e^{-ca})$, small only when ca is large relative to $\ln(K-1)$; at $a \rightarrow 0$ the anchors merge onto a single origin and FGIB degenerates to a VIB-style marginal prior (Remark 22). Table 3 holds β at the largest grid value — maximum compression pressure — and varies a .

The prediction holds. At $a = 1$, FGIB behaves like the marginal-prior bottlenecks: 1.0% on ImageNet-100/MLP (chance for 100 classes), 26.6% on Cora (below the majority class), $R^2 = -0.001$ on AgeDB/MLP (the mean predictor). Growing a recovers the backbone — monotonically on ImageNet-100/MLP ($1.0 \rightarrow \dots \rightarrow 83.0$, 0.2 below its backbone), with local dips elsewhere (Cora ends 87.2%, AgeDB/MLP $R^2 = 0.293$, 7.5 points *above* its backbone). The trend, not the monotonicity, is what the theory predicts, and it holds on 10 of the 12 rows. The two rows with no upward trend are the two the theory says should not have one: IMDB is binary, so the corner value is already negligible at $a = 1$ (the whole row sits above its backbone at every a); ZINC is a regression row, where the random-Fourier anchors carry no corresponding guarantee (Remark 16) and it stays 4–6 points below its GCN throughout. Consistent with the $(K-1)$ factor, the 100-class task needs by far the largest anchor (MNIST/MLP, $K=10$, is within 0.3 points at $a = 1$; ImageNet-100/MLP needs $a \geq 10$).

The formula does not predict the transition threshold, and we should not pretend it does. Cora ($K=7$) collapses at $a \leq 2$ where MNIST ($K=10$) does not, so K is not the only factor — the classifier scale c enters the corner value multiplicatively and is free in our implementation (Remark 12), and Cora is a small transductive problem with correspondingly different margins. What Table 3 establishes is the qualitative mechanism — recovery with a , chance-level collapse as $a \rightarrow$ the VIB limit, largest- K needs largest a — not a quantitative fit, and not strict monotonicity.

4.5 CONFOUND ABLATIONS AND DIAGNOSTICS ON MNIST

Section 3.3 identifies the channels FGIB leaves open; this section reports the supplementary experiments that test whether they bite, that separate FGIB’s two design changes from each other, and that verify the two closure results of Section 3. All runs use the identical protocol as the main sweep (100 epochs, patience 10, 5 seeds), except the long-horizon control noted below; the addi-

tional experiments total 360 runs across a 12-setting CentFGIB sweep at equal budget, a 4-setting FGIB-H ($A=I$) sweep, a full-batch diagnostic rerun, a 400-epoch long-horizon control, a frozen-ETF baseline, and a subject-disjoint AgeDB sweep. Figure 1 plots the full β sweeps and Table 10 the numbers.

What the ablations show. FGIB differs from the comparison in two coupled ways: its classifier consumes the deterministic h at training and evaluation (P3, P4), and its prior is frozen (P1, P2). Table 10 separates the contributions. VIB and CEB fall to chance at $\beta = 1$ (15.3% and 26.8%); *DCEB* — the deterministic path of FGIB with CEB’s trainable backward encoder as its prior — holds at ≈ 98 through $\beta = 1$ and falls to 11.4% only at $\beta = 10$. Corollary 1 explains *why* DCEB is the odd one out: its trainable prior *variance* has a negative gradient in the mean gap, so the race of Channel P3 re-enters through the prior — the only FGIB-family variant that escapes the exact center-loss family is the only one that collapses. By contrast, every variant that *starts* from the orthonormal anchor table stays at ≈ 98.3 across the entire grid (TAFGIB, CentFGIB, frozen A , FGIB-H) — and by Result 2 that equivalence is an identity, not a coincidence: the zero-initialized variance head receives zero gradient at every step, so FGIB’s KL is the center loss exactly, and CentFGIB, TAFGIB and FGIB-H are the same objective up to the trainability of the anchor means. The trainable anchors of TAFGIB do move (final column norms shrink from 16 to 5.7–10), yet the accuracy curve is untouched — the load-bearing ingredient is the anchor-table *initialization* geometry at scale a , not the frozenness of its means and not any Gaussian variance machinery, which is provably inert here (Result 2). What freezing the means and keeping the KL buys is the *theory*: the persistent force of Remark 17, the convexified sweep of Proposition 8, and the residual-information reading of Result 1 — FGIB retains an exact variational upper-bound interpretation for the auxiliary noisy code; its predictive robustness outside the analyzed family remains empirical. Constraining the side head changes nothing in the β -profile on MNIST (98.2–98.4 across the grid for frozen A and $A=I$), and the fixed classifier scale test agrees: enforcing the cosine condition leaves FGIB’s sweep unchanged (97.9–98.3) while VIB and CEB still collapse, so the anchor geometry, not the free classifier scale, keeps the corner non-degenerate.

The variance race is real but slow. Logging the mean posterior log-variance, the clamp fraction (e^{-10}), and (for CEB) the prior log-variance at every epoch of every run, aggregated over the *entire* validation set, CEB’s final posterior log-variance sits at +0.37, +0.27, +0.13, +0.08, −0.05, −0.01 (nats) for $\beta = 10^{-4}, \dots, 10$ — near initialization, nowhere near the clamp, clamp fraction 0.0 at every epoch (Figure 2(b)). A 400-epoch control at $\beta \in \{1, 10\}$ (early stopping disabled) shows the race’s *direction*: the posterior log-variance ends at −2.27 nats ($\beta=1$) and −1.20 ($\beta=10$), passing −0.92 at epoch 100, the prior tracking — monotone, exactly as Proposition 3(i)–(ii) predicts, but still far from the e^{-10} terminal state (Figure 3). The race is therefore real but slow: invisible at the protocol horizon, directionally confirmed at four times it, its convergence time-scale beyond the tested 400 epochs — the conditional status Remark 4 assigns to part (iii). For FGIB the question is settled analytically: the flat zero line of Figure 3 is the fixed point of Result 2, held for 400 epochs, with the trainable prior variance of DCEB the only re-entry point (Corollary 1).

The side head: measured conditioning, quantified closure. The trainable head’s $\log_{10} \kappa(A^\top A)$ rises from $10^{3.9}$ at initialization to $10^{5.8}$ – $10^{6.8}$ (transient peaks $10^{8.3}$; Figure 2(a)), while accuracy is unaffected on MNIST. The canonical FGIB-H ($A=I$) closes the channel by construction (Result 3), and Table 8 quantifies the price across five settings: at the inherited scale $a = 16$ FGIB-H stays ahead of the backbone on 4 of 5 settings but trails the trainable head by ≈ 0.5 points per setting; at the re-tuned scale $a = 4$ it recovers most of the gap (mean rank 2.25 vs. 1.20) and ties or beats the backbone on every row — the trade-off is accuracy-versus-certification, small in both directions, and both configurations ship in the released code. The cross-dataset reach of the geometry itself is measured separately: CentFGIB matches FGIB($a=16$) to within 0.01 points on 10 of 12 settings at equal budget (Table 7), a frozen ETF classifier alone reproduces the backbone but buys no knob (98.32% vs. 98.33%), and a subject-disjoint AgeDB split preserves the objective ordering (Table 9). The full details of this section — per-variant grid tables, the TAFGIB geometry measurements, the fixed-scale rows, and the anchor-scale sensitivity of the equal-budget rank — are in Appendix E. The full per-setting anchor dial (Table 3) and the β -sweep curves (Figure 1) are likewise in Appendix D.

4.6 WHAT WE HAVE NOT MEASURED

The experiments this paper’s own theory originally asked for are now in it (Section 4.5): the variance-race measurement (on the full validation set, out to 400 epochs), the $\kappa(A^\top A)$ conditioning measurement, the fixed-orthogonal closure (Table 8), and a subject-disjoint AgeDB control (Table 9). What remains open is on the record in Appendix F.

5 RELATED WORK

Variational bottlenecks. VIB (Alemi et al., 2017) bounds $I(X; Z)$ with a marginal prior; CEB (Fischer, 2020) bounds the residual $I(X; Z|Y)$ with a learned backward encoder and is tighter by exactly $I(Y; Z)$ at the respective optima. FGIB inherits the residual-level object and replaces the learned backward encoder with a frozen geometric one; Proposition 10 shows the resulting bound is looser in value by the anchor gap, and Remark 17 argues that this is what makes its force persistent. DVCCA (Abdelaleem et al., 2025) sits in the same variational family with an added reconstruction term, and inherits the marginal-prior corner.

Degeneracies of the IB Lagrangian. Kolchinsky et al. (2019a) prove that in the deterministic scenario the IB curve is piecewise linear, so the Lagrangian cannot sweep it, and repair this by squaring the compression term. Our Proposition 1 shows CEB inherits the degeneracy over its entire range, and Result 4 shows that a fixed reference supplies the same strict convexity *without* squaring — while squaring alone (SVIB) does not fix the collapse (Table 2). Kolchinsky et al. (2019b) avoid the parametric prior entirely with a non-parametric pairwise bound; our experiments locate the price of that route in an inert knob.

Deterministic representations and infinite information. Amjad & Geiger (2020) and Kolchinsky et al. (2019b) observe that $I(X; Z)$ is infinite or ill-defined for deterministic continuous encoders, and Saxe et al. (2018) question the empirical IB narrative on related grounds; sample-based limits on measuring mutual information (McAllester & Stratos, 2020; Tschannen et al., 2020) apply to any bound-as-meter reading, ours included. FGIB’s response is architectural: keep the inference path deterministic and confine the noise — and thus the finite bound — to a side path that inference never reads.

Fixed geometric targets. Center loss pulls features toward learned per-class centers (Wen et al., 2016); hyperspherical prototype networks place prototypes on a hypersphere a priori (Mettes et al., 2019); the final classifier itself can be fixed with little accuracy loss (Hoffer et al., 2018), in particular to a simplex equiangular tight frame (Zhu et al., 2021; Pernici et al., 2019); and neural-collapse analyses show such configurations arise spontaneously (Papayan et al., 2020). FGIB’s compressed corner is a neural-collapse-like geometry — orthonormal class codes rather than the centered simplex/ETF configuration — but the construction differs in the role the geometry plays: it enters as the *reference of a variational bound*, i.e. as the design of the bottleneck’s zero-penalty set, which makes the compression term quadratic (Result 4) and gives the regularizer a residual-information reading for the auxiliary code (Result 1) rather than serving as an auxiliary classification loss (Table 11, Appendix D). The boundary of that claim is a theorem rather than an admission: in the released implementation the KL is the fixed-anchor center loss (Result 2), and the canonical fixed-orthogonal head moves the matching onto the deployed representation itself (Result 3) — the *empirical* robustness is shared with fixed-prototype regularization by construction, while the bound reading and the sweep theory are specific to the variational form. The natural baselines from this literature are tested directly: a frozen ETF classifier alone reproduces the backbone but buys no knob, and trainable-center and fixed-center losses are the TAFGIB and CentFGIB rows of the ablation (Section 4.5).

6 CONCLUSION

The high-regularization behavior of a variational bottleneck is decided by its zero-penalty set, and the standard choices are poorly designed: VIB’s marginal prior places that set at collapse, CEB’s learned backward encoder leaves it underdetermined, and both certify the noisy training code rather than the deterministic representation inference deploys (a gradient-level variance race, in contrast, is

invisible at the protocol horizon and only drifts into view at 400 epochs). Designing the set directly — an orthonormal anchor frame fixed by construction — makes the compression term quadratic, well-poses the sweep at fixed classifier scale, and turns the fully compressed corner into the prescribed class-separated anchor configuration. Two properties of the construction are provable rather than merely observed: the deterministic main path pins the variance at τ^2 for the entire run, so the realized regularizer is *exactly* the fixed-anchor center loss $\frac{1}{2\tau^2}\|\mu - av_y\|^2$ — releasing the variance is what makes the alternatives race to the clamp, and FGIB eliminates the race by design; and the canonical fixed-orthogonal side head anchors the deployed representation directly with $\kappa = 1$, closing the conditioning channel a trainable head leaves open, at a measured price of ≈ 0.5 points per setting that re-tuning the anchor scale largely recovers. Empirically, across 12 task×backbone settings and 5,100 runs, that is worth a first-place mean rank under the equal-budget configuration (2.33 over all 12; 1.75 over the full grid; 1.86 classification-only; 1.70 excluding AgeDB) and, more distinctively, a compression knob that can be swept over a six-point grid spanning five orders of magnitude without destroying the model on classification — something no other objective in the comparison manages. The ablations pin down what carries that robustness: the endpoint geometry, not any Gaussian variance machinery — which is provably inert here — not the frozenness of the anchor means, and not the deterministic path alone; the one variant that re-opens the variance, DCEB, is the one that collapses. The KL’s role is interpretive: an exact variational upper bound on the residual information of the auxiliary noisy code, the persistent force and convexified sweep of the theory. The cross-dataset reach of the geometry-only mechanism is measured — CentFGIB matches FGIB at equal budget on all 12 settings, FGIB-H quantifies the fixed head’s price across five (Section 4.5) — and a subject-disjoint AgeDB control preserves the regression ordering. What remains on the ledger: split-sensitivity beyond AgeDB, budget expansion for the comparison methods, and the downstream robustness properties that motivate bottlenecks in the first place.

AI USE STATEMENT

(This section is **required** and does not count toward the page limit.)

In this work, we used generative AI tools (large-language-model assistants) for code generation and refactoring, drafting and editing portions of the text, and checking mathematical derivations. All AI-assisted work was reviewed and verified by the authors, including every equation, table number and experimental claim. We take responsibility for the final content of this work, including text, claims or artifacts produced with the aid of generative AI. See the ICLR 2027 AI Policy for Authors for more details.

ETHICS STATEMENT

The datasets used are standard public benchmarks. One of them, AgeDB (Moschoglou et al., 2017), consists of images of human faces annotated with age, and we use it only as a scalar regression benchmark; we do not train or release face recognition or demographic-inference models, and we make no claim that age estimates from this data generalize fairly across populations. No new data was collected and no human subjects were involved.

REPRODUCIBILITY STATEMENT

All results come from a single training entry point with a shared argument parser across tasks, backbones and objectives; the exact grid is described in Section 4 and every one of the 1,020 configurations writes a log containing the per-run and averaged test metrics. Tables 1–4 are generated from those logs by a single script (`make_tables.py`) that re-derives the validation selection, so no number in this paper is transcribed by hand; the paired statistics of Table 5 (`make_stats.py`) and the ablation table of Table 10 (`make_ablation.py`) are generated the same way. All proofs are in Appendix A, with the assumptions of each statement made explicit and the unproved items collected in a single scope remark. Data preprocessing, splits (fixed seed 42) and caching are described in Appendix G. Model selection uses validation scores only — every tie-break is a second validation quantity or fixed grid order — and test scores are read once per configuration.

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A COMPLETE THEORY: PROOFS OF THE FAILURE CHANNELS AND OF THE FGIB MECHANISM

The remainder of this appendix is the full development: Part I proves the three failure channels and the object mismatch of Section 2, Part II proves the FGIB results of Section 3, and its final subsection (Section A.4) is a complete ledger separating what is proved from what is open or conditional.

PART I. THREE FAILURE CHANNELS OF THE CONDITIONAL BOTTLENECK AND AN OBJECT MISMATCH

Notation. X is the input and Y the label with K classes; the encoder induces the Markov chain $Y - X - Z$. The *deterministic scenario* is $Y = f(X)$ ($H(Y|X) = 0$). Following Fischer (2020), we write the MNI point $I(X; Y) = I(X; Z) = I(Y; Z)$ (Eq. 1), the chain-rule identity $I(X; Z|Y) = I(X; Z) - I(Y; Z)$ (Eq. 4), the CEB objective $\text{CEB} \equiv \min_Z I(X; Z|Y) - \gamma I(Y; Z)$ (Eq. 5), the IB objective $\text{IB} \equiv \min_Z I(X; Z) - \beta_{\text{IB}} I(Y; Z)$ (Eq. 16), the variational objective $\text{VCEB} \equiv \min_{e,b,c} \langle \log e(z|x) \rangle - \langle \log b(z|y) \rangle - \gamma \langle \log c(y|z) \rangle$ (Eq. 15), and the residual certificate $\text{ReX} \equiv \langle \log e(z|x) \rangle - \langle \log b(z|y) \rangle \geq I(X; Z|Y)$ (Eq. 20), where $b(z|y)$ is a *trainable* backward encoder (we write $q(z|x)$ for Fischer’s encoder $e(z|x)$; the two notations are interchangeable throughout). In the loss convention $\text{CE} + \beta \text{KL}$ used throughout, the weights are $\beta = 1/\gamma$ and $\beta_{\text{IB}} = 1 + \gamma = 1 + 1/\beta$. Following Kolchinsky & Tracey (2019), we write the IB curve $F(r)$ (their Eq. 1), the IB Lagrangian $L_\beta^{\text{IB}} = I(Y; Z) - \beta I(X; Z)$ (their Eq. 2), the DPI-saturating manifold $T_\alpha := B_\alpha \cdot Y$ (their Eq. 6), the bound $L_\beta^{\text{IB}} \leq (1 - \beta)H(Y)$ (their Eq. 7), and their footnote 4 (the $\beta > 1$ regime of trivial maximizers). The variational objects are

$$R = \mathbb{E}[\text{KL}(q(z|x) \| m(z))] \geq I(X; Z) \quad (\text{VIB rate, marginal prior } m; \text{Fischer Eq. 19, Alemi et al. Eq. 14})$$

$$\text{ReX} = \mathbb{E}[\text{KL}(q(z|x) \| b_\theta(z|y))] \geq I(X; Z|Y) \quad (\text{CEB residual; Fischer Eq. 20})$$

$$\hat{I} = -\frac{1}{N} \sum_i \ln \left[\frac{1}{N} \sum_j e^{-\|f_i - f_j\|^2 / (2\sigma^2)} \right] \geq I(X; M) \quad (\text{NIB, non-parametric; Eq. 16, the homoscedastic case of Eq. 10})$$

$$\text{ReX}_F = \mathbb{E}[\text{KL}(q(z|x) \| r(z|y))] \geq I(X; Z|Y) \quad (\text{FGIB, fixed anchor})$$

(Fischer 2020, Eqs. 18–20; Alemi et al. 2017, Eq. 14; Kolchinsky et al. 2017, Eqs. 10 and 16). Fischer’s comparison of the first two: at the respective prior optima ($m = q(z)$, $b = q(z|y)$), $\text{ReX} = R - I(Y; Z)$ — the conditional prior removes exactly the retained label information, so CEB’s bound is tighter than VIB’s by $I(Y; Z)$. We ask the same question one level down: where does FGIB’s fixed-anchor bound sit relative to CEB’s learnable-backward bound?

Proposition 1 (Channel P1, dead knob: every $\gamma > 0$ selects the same corner). *Assume $Y = f(X)$. For every $\gamma > 0$,*

$$\min_Z L_\gamma(Z) = -\gamma H(Y), \quad L_\gamma(Z) := I(X; Z|Y) - \gamma I(Y; Z), \quad (5)$$

attained exactly on the corner family $\{Z : I(Y; Z) = H(Y), I(X; Z|Y) = 0\}$ (e.g. $Z = Y$, or $Z = (Y, N)$ with $N \perp X | Y$): the corner is unique in the information plane, while at the representation level it is an entire family. Consequently the map $\gamma \mapsto (I(X; Z^|Y), I(Y; Z^*))$ is constant: no interior point of the IB curve is ever a Lagrangian minimizer, and CEB inherits the deterministic degeneration of the Caveats paper (their Issue 1) over its entire parameter range, since $\beta_{\text{IB}} = 1 + \gamma > 1$ always.*

Proof. By Fischer’s Eq. 4, $L_\gamma(Z) = I(X; Z) - (1 + \gamma)I(Y; Z)$. The DPI ($I(Y; Z) \leq I(X; Z)$), by the Markov chain $Y - X - Z$) gives

$$L_\gamma(Z) \geq I(Y; Z) - (1 + \gamma)I(Y; Z) = -\gamma I(Y; Z) \geq -\gamma H(Y), \quad (6)$$

with equality iff both inequalities bind: $I(X; Z) = I(Y; Z)$ and $I(Y; Z) = H(Y)$. $Z = Y$ attains the bound. Fischer (2020, p. 7) states the underlying equivalence explicitly: CEB and IB are the same objective for $\gamma = \beta_{\text{IB}} - 1$. In the Caveats paper’s maximizing convention, minimizing L_γ is maximizing L^{IB} with $\beta' = 1/\beta_{\text{IB}} = 1/(1 + \gamma) \in (0, 1)$ — precisely the range in which their Eq. 7 pins every maximizer at the copy corner (their Issue 1); their footnote 4 covers the complementary regime $\beta' > 1$ (i.e. $\gamma < 0$, outside CEB’s range), whose maximizers are the trivial collapse. $\square$ $\square$

Remark 1 (The corner is the MNI point: why that does not rescue the knob). *In the deterministic scenario the corner of Proposition 1 is the MNI point: $I(Y; Z) = H(Y) = I(X; Y)$ together with $I(X; Z|Y) = 0$ gives $I(X; Y) = I(X; Z) = I(Y; Z)$, which is Fischer’s Eq. 1. CEB’s declared target and the degenerate attractor therefore coincide, and Kolchinsky & Tracey concede the point themselves (their Section 4): “if $Y = f(X)$ and one’s goal is precisely to find the corner point $(H(Y), H(Y))$ (maximum compression at no prediction loss), then our results suggest that the IB*

Lagrangian provides a robust way to do so, being invariant to the particular choice of β .” Read that way, Proposition 1 states hyperparameter robustness rather than failure. Three things keep it from closing the matter.

- (i) The corner is one point in the information plane and an entire equivalence class of representations. *The objective is exactly constant on $\{Z : I(Y; Z) = H(Y), I(X; Z|Y) = 0\}$, which contains $Z = Y$, $Z = (Y, N)$ with $N \perp X | Y$, and every bijective recoding of these. Adversarial robustness, OoD detection and calibration — the properties CEB is argued for — are properties of the geometry of Z , on which the objective has no opinion; which member of the class is reached is settled by the optimizer. That is channels P3 and P4 below.*
- (ii) CEB’s own reading of the knob is unavailable at the exact optimum. *Fischer (2020, p. 7) writes “ $\rho < 0$ may capture less than the MNI. $\rho > 0$ may capture more”, whereas Proposition 1 places every $\gamma > 0$ at the MNI point exactly. His argument for preferring $\gamma = 1$ is correspondingly not about minimizers but about the optimizer: weighting the two appearances of $I(Y; Z)$ differently “forces the optimization procedure to count bits of $I(Y; Z)$ in two different ways” (his Eq. 8 and surrounding discussion).*
- (iii) The knob does move trained models, which locates what it acts on. *On Fashion MNIST — a deterministically labelled dataset — Fischer’s own Figure 4 reports the rate lower bound R_X rising from ≈ 2.0 to ≈ 5.0 nats as ρ goes from -1 to 5 , with $\rho = 0$ landing on $I(X; Y) = \log 10 \approx 2.3$ nats. Since every $\gamma > 0$ shares one exact optimum, that axis is not a path along the IB curve: it is the distance between the optimum of the variational objective over a restricted family and the point at which training was stopped. A knob acting through the approximation error is precisely the knob that Propositions 2 and 3 show cannot be read off the certificate.*

Kolchinsky & Tracey state that their results “are based on analytically-provable properties of the IB curve, i.e., global optima” and “do not concern practical issues of optimization”; Proposition 2 supplies the finite- β variational counterpart. The account is owed by any method reporting a β -sweep on deterministically labelled data — FGIB and the experiments of this paper included (Remark 12).

Proposition 2 (Channel P2, a blind certificate and a $1:\beta$ exchange rate). *On the manifold $T_\alpha = B_\alpha \cdot Y$ of the Caveats paper (their Eq. 6), $I(X; T_\alpha|Y) = 0$ for every α , and with the optimal backward encoder and classifier,*

$$\text{ReX}(T_\alpha) = 0, \quad \text{CE}(T_\alpha) = H(Y) - I(Y; T_\alpha) =: \delta_\alpha, \quad \text{VCEB}(T_\alpha) = \gamma \delta_\alpha. \quad (7)$$

Two consequences. (a) The certificate is blind to label retention. $\text{ReX} = 0$ holds from the total collapse ($\alpha = 0$, no label information) to the full copy ($\alpha = 1$), so its tightness is compatible with retaining all, part, or none of Y . Fischer’s reading of the certificate as a meter — “observing ReX tells us exactly how many bits we are from the MNI point” (p. 7) — therefore holds only along the $I(X; Z|Y)$ axis; on the necessity axis the meter reads 0 at chance level. (b) Exchange rate. With the optimal classifier the objective is $L = \text{CE} + \beta \text{ReX} = \delta + \beta \text{ReX}$, so it is indifferent between giving up δ nats of label information and saving δ/β nats of certified residual. In the heavy-regularization regime the bottleneck is supposed to act in, the whole label axis — up to $H(Y)$ nats — is purchasable for $H(Y)/\beta$ nats of certified compression.

Proof. T_α is a function of Y alone, so $T_\alpha \perp X | Y$ and $I(X; T_\alpha|Y) = 0$; the Caveats paper shows $I(Y; T_\alpha)$ sweeps $[0, H(Y)]$. With the true backward encoder ($b = p(z|y)$) the certificate is tight: $\text{ReX} = I(X; T_\alpha|Y) = 0$ (Fischer Eqs. 9–11, 20). The optimal classifier achieves $\langle \log c(y|z) \rangle = -H(Y|T_\alpha)$ (Fischer Eq. 12), i.e. $\text{CE} = H(Y|T_\alpha) = \delta_\alpha$, and the VCEB value follows from Eq. 15. Part (b) is the definition of the objective evaluated at $\text{CE} = \delta$. $\square$ $\square$

Remark 2 (What the flatness claim is, and what it is not). *An earlier form of this argument read the near-optimality band off the value $\text{VCEB} = \gamma\delta \rightarrow 0$. That reading is a scaling artifact: $\text{VCEB} = \gamma(\text{CE} + \beta \text{ReX}) = \gamma L$, so the two conventions differ by a positive factor and share their minimizers, and on T_α itself the objective is $L = \delta$ in either convention and strictly prefers $\alpha = 1$. What is convention-free is the ratio of Proposition 2(b): the prediction term is weaker than the compression term by exactly the factor β , so in units of the compression term the entire manifold lies*

within $H(Y)/\beta$ of optimal. Under a scale-invariant optimizer (Adam, used throughout the experiments) only the ratio matters; under plain SGD at fixed learning rate the convention additionally rescales the effective step size. Combined with Proposition 2(a) — no certified compression is bought anywhere on the manifold — the operative statement is that the label axis is cheap and unmetered, not that the objective is literally flat along it. Pushing β up also drives $\beta_{\text{IB}} = 1 + 1/\beta \rightarrow 1^+$: the objective approaches the Caveats paper’s $\beta = 1$ degeneracy (their Eq. 7 discussion), at which every point of the diagonal is simultaneously optimal, from above. Fischer (2020, p. 7) anticipates the direction — “ $\rho < 0$ may capture less than the MNI” — but by Proposition 1 that cannot happen at the exact optimum; Proposition 2 gives the rate at which the variational objective is willing to let it happen.

Remark 3 (Approximately deterministic labels). *The Caveats paper (Appendix C, Issue 1) shows that when Y is only ϵ -close to deterministic, all optimizers fall within $O(-\epsilon \log \epsilon)$ of a single corner: the dead-knob statement of Proposition 1 degrades gracefully, and the near-optimality band of Proposition 2 is perturbed by the same order.*

Proposition 3 (Channel P3, collusion collapse: the variance race and the overfit channel). *For the diagonal Gaussian family $e(z|x) = \mathcal{N}(m(x), \sigma^2 I)$, $b(z|y) = \mathcal{N}(g(y), \tau^2 I)$ (the implementation’s setup, with the logvar clamp $[-10, 10]$ of `kl-divergence`), and a linear classifier $c(z) = Wz + b$ (the implementation’s classifier is `nn.Linear`):*

- (i) *for fixed means, KL is minimized over σ^2 at $\sigma^2 = \tau^2$, with value $\frac{1}{2\tau^2} \|m - g\|^2$ — a pure mean-matching term;*
- (ii) *the classifier term favors $\sigma^2 \rightarrow 0$: $\text{CE}(c(z), y) = -\log \text{softmax}_y(Wz + b)$ is convex in z (log-sum-exp of an affine function of z , minus a linear term — the convexity claim requires c affine), so Jensen gives $\mathbb{E}_z[\text{CE}(c(z), y)] \geq \text{CE}(c(m), y)$. The KL’s derivative in σ^2 vanishes at $\sigma^2 = \tau^2$, but along the matched line it resists shrinking τ^2 while the means are unmatched ($\partial \text{KL} / \partial \tau^2|_{\sigma^2=\tau^2} = -\|m - g\|^2 / (2\tau^4) < 0$). The joint dynamics therefore runs in two stages: the mean matching $m \rightarrow g$ completes first (with weight $1/\tau^2$, which grows as τ shrinks), and once $\|m - g\| \rightarrow 0$ the variance race is unopposed — the classifier drives $\sigma^2 \rightarrow 0$ with τ^2 following the moment-matched value — until the clamp at $s = e^{-10}$, where the encoder collapses toward the deterministic code $z \approx g(y)$ and $\text{ReX} = \frac{1}{2s} \mathbb{E} \|m(x) - g(y)\|^2$: a mean-matching force of weight $1/s = e^{10} \approx 2.2 \times 10^4$ between two trainable maps;*
- (iii) *the certified object and the deployed object separate. By the gap lemma (Lemma 4) $\text{ReX} = I(X; Z|Y) + \Delta \geq I(X; Z|Y)$, so a small ReX entails a small residual for the stochastic training code — the certificate cannot under-report its own object. What it does not control is the relation between the training code and the deployed representation. Exact mean matching ($m(x) = g(y)$) makes both objects class-constant, so both residuals vanish; the scenario of interest is the near-match regime, where $\mathbb{E} \|m(X) - g(Y)\|^2 = o(s)$ keeps ReX small while the deployed continuous m retains within-class variation, and any non-class-constant continuous m has $I(X; m(X)|Y) = +\infty$ under the standing non-atomic assumption (the pathology of deterministic networks, Amjad & Geiger 2018; Kolchinsky et al. 2017). The variance race therefore moves the certified object toward a noise-dominated class-constant code while the deployed object can keep diverging residual — and the encoder is fitted to the backward encoder’s per-class targets, memorizing the label code $g(y)$ rather than being told what residual to erase, while the certificate reports $\text{ReX} \approx 0$ on a code that inference does not use.*

Proof. The closed form $\text{KL} = \frac{d}{2} [\sigma^2/\tau^2 + \|m - g\|^2 / (d\tau^2) - 1 + \ln(\tau^2/\sigma^2)]$ has derivative $\frac{d}{2} (1/\tau^2 - 1/\sigma^2)$ in σ^2 , so the variance optimum is $\sigma^2 = \tau^2$, giving (i); at $\sigma^2 = \tau^2 = s$ the value is $\|m - g\|^2 / (2s)$. For (ii): $\text{CE}(c(z), y)$ has Hessian $\text{diag}(p) - pp^\top \succeq 0$ (the categorical Fisher information), so it is convex in z and Jensen gives the stated inequality; the same Hessian identity makes $\sigma^2 \mapsto \mathbb{E}_z[\text{CE}(c(z), y)]$ non-decreasing (Gaussians of larger variance dominate in convex order), strictly increasing in σ^2 wherever the classifier is unsaturated, so that term pushes σ^2 down. The KL’s partial derivatives at the matched point are $\partial_{\sigma^2} \text{KL} = 0$ and $\partial_{\tau^2} \text{KL} = -\|m - g\|^2 / (2\tau^4) < 0$, so the KL pulls τ^2 up until the means match, while the mean-matching gradient on the encoder is $(m - g)/\tau^2$; once $\|m - g\| \rightarrow 0$ the race to the clamp is unopposed. For (iii): the gap identity (Lemma 4) forces the certified residual and the certificate to vanish together; the divergence attaches

to the deployed mean $m(x)$ whenever it is not class-constant (Amjad & Geiger; Kolchinsky et al.).

□

□

Remark 4 (Status of the two-stage account). *Parts (ii) and (iii) are a stationarity-and-direction-of-force argument, not a convergence proof: the “two stages” presume a time-scale separation (mean matching completes before the variance race) that is not established here. The account is directly falsifiable in this codebase: log the mean posterior log-variance and the fraction of units at the clamp $s = e^{-10}$ during CEB training. We ran the test, on the full validation set of every epoch (aggregated over all batches; Section 4.5 of the main paper). On MNIST, across all six β settings and 5 seeds, the trained posterior and prior log-variances of CEB remain near their initialization ($\sigma^2 = \tau^2 = 1$, mean $\log \sigma^2$ within 0.74 of zero, clamp fraction 0.0 at every epoch of every run). A 400-epoch control at $\beta \in \{1, 10\}$ (no early stopping, Section 4.5) resolves the status more precisely: the variance does drift, monotonically and in exactly the direction (i)–(ii) predict — posterior $\log \sigma^2$ ends at -2.27 nats ($\beta = 1$) and -1.20 ($\beta = 10$), prior tracking — but the clamp fraction remains 0.0 and -2.3 nats is far from the e^{-10} terminal state of part (iii). The race is therefore real but slow: invisible at the protocol horizon, directionally confirmed at four times it, convergence beyond the tested 400 epochs. That is precisely the conditional status this remark assigns to Proposition 3(iii): a direction-of-force statement whose terminal state the longest available run has not reached, and the measurement itself — not the prediction — as the finding. On the FGIB side the question does not arise as an empirical one at all: Proposition 7 proves the posterior variance is pinned at τ^2 by construction — the flat zero line of the 400-epoch run — and Corollary 1 shows the race can re-enter only through a trainable prior log-variance — DCEB’s configuration — whose measured collapse at $\beta = 10$ is consistent with that channel.*

Remark 5 (Train–evaluation mismatch). *The classifier is trained on the sampled code z and evaluated on the mean $m(x)$ (Fischer 2020, Appendix A.1). Proposition 3(ii) is the force that closes this gap by destroying the noise altogether ($\sigma \rightarrow 0$), at which point the bottleneck’s stochasticity — the object the bound is defined on — disappears.*

Proposition 4 (Channel P4, certificate-object mismatch: neither bound is an upper bound on the deployed representation). *(a) CEB. The classifier consumes the sampled code Z at training and the mean $\mu(X)$ at evaluation (Fischer 2020, Appendix A.1). At the variance-optimal posterior, $\sigma(x) \equiv \sqrt{S_\theta(y)}$ (the backward encoder’s per-class standard deviation) is class-constant, so $Z = \mu(X) + \sqrt{S_\theta(Y)} \varepsilon$ is a noisy function of $\mu(X)$ given Y ; by the DPI,*

$$I(X; Z|Y) \leq I(X; \mu(X)|Y), \quad (8)$$

strictly so whenever the noise destroys within-class detail. Hence CEB’s certificate $\text{ReX} \geq I(X; Z|Y)$ is a bound on the training code, not on the residual of the representation deployed at inference. The gap is not a small correction: whenever μ is a continuous map that is not class-constant, $I(X; \mu(X)|Y) = +\infty$ under the standing non-atomic assumption (Amjad & Geiger 2020; Kolchinsky et al. 2017), so no finite quantity — CEB’s certificate included — bounds the deployed residual from above. The CE term presses the noise toward 0 (by Jensen, $\mathbb{E}_z[\text{CE}(c(z), y)] \geq \text{CE}(c(\mu), y)$), but no finite σ closes the gap. (b) FGIB. The inference path is h at both training and evaluation; the side noise of Proposition 11 is not part of any inference path. By the DPI applied to the same perturbed-variable construction,

$$I(X; h + \tau(A^\top A)^{-1/2} \varepsilon | Y) \leq I(X; H|Y), \quad (9)$$

so the direction of the bound is the same as CEB’s: ReX_F upper-bounds the noisy side-path object, and no finite value upper-bounds the deployed residual. The point-anchor limit $\tau \rightarrow 0$ does not repair this: it is a change of model whose bound value $\|\mu - av_y\|^2/(2\tau^2)$ typically diverges (Remark 10), and even the limiting object needs A well-conditioned (Proposition 12(iii)). (c) Conclusion. In the pure variational value sense (same encoder, same Gaussian family), CEB’s bound is tighter (Proposition 10). In the object sense the two are not ordered: both are valid upper bounds on their respective noisy codes, and neither bounds the deterministic deployed representation. What remains structural — not a tightness claim — is where the gap lives: CEB’s perturbation is inside the inference path and is driven by the prediction term (so removing it changes the model), while FGIB’s perturbation lives on a side path that inference never reads and is controlled by side-path parameters alone. The latter is a cleaner separation of regularizer from model, not a tighter bound.

Remark 6 (The bound is a training instrument, not an inference one). *No method computes its bound at inference: the bound’s roles are training-time — a regularizer (a differentiable surrogate for the*

intractable information term) and a meter (Fischer 2020, p. 7: “observing ReX tells us exactly how many bits we are from the MNI point”). In the $Z \perp X \mid Y$ limit CEB’s mean is also class-constant ($m(x) = \mathbb{E}[Z|x] = \mathbb{E}[Z|y]$), so the mismatch of Proposition 4(a) is not a limit failure: it is a finite- β under-reporting — the regularizer controls, and the meter reports, the training code, which is strictly more compressed than the deployed representation $\mu(X)$, by a positive, trained deficit that cannot be measured from the certificate. FGIB’s side-path noise is a known side-path parameter and is removable in principle ($\tau \rightarrow 0$ changes the side path, never the deployed model), but for finite τ its regularizer and meter act on a perturbed copy of the deployed object, with the perturbation size governed by the conditioning of A (Proposition 12(iii)). The one genuine inference-time use of a certificate value is as a detection score (Fischer 2020, RG3, uses the marginal rate R): CEB’s label-conditioned ReX is not even computable at inference (the label is unknown — Fischer trains a separate marginal for this purpose), while FGIB’s geometry gives a label-free score, the distance to the nearest anchor $\min_y \text{KL}(q(z|x) \parallel \mathcal{N}(av_y, \tau^2 I))$, with no extra network (the minimum is an internal enumeration over the K fixed anchor buffers — the single QR orthonormal frame — and consults no ground-truth label, no more than softmax consults labels when it enumerates the K logits). Whether this score is useful for OoD detection is a hypothesis; nothing here proves it.

The problem addressed by FGIB. The channels are addressed by the FGIB design as follows; two are closed by construction, one holds conditionally, and the mismatch is only partially addressed. The precise scope of each claim is documented in Part II and the open items are collected in Section A.4.

- *Dead knob and cheap label axis* (Props. 1, 2): FGIB’s compression term is the anchor gap $\frac{1}{2\tau^2}(s - a)^2$, quadratic in the alignment scalar, and at fixed classifier scale the Lagrangian $\text{CE}(s) + \frac{\beta}{2\tau^2}(s - a)^2$ is strictly convex with a unique interior minimizer per β — the β -sweep is one-to-one onto the whole trade-off curve (Props. 8, 9). With the scale free the infimum is 0 for every β ; Remark 12 states the condition and the architectural fix.
- *Blind certificate* (Prop. 2): with the fixed anchor $r(z|y) = \mathcal{N}(av_y, \tau^2 I)$ the gap $\mathbb{E}_y[\text{KL}(q(z|y) \parallel r(z|y))]$ can shrink only if the *posterior* moves toward the anchors — the reference itself cannot chase the encoder, so the force is persistent (Prop. 10 and Remark 17). The side head A , however, is a second trainable party in that matching, and it is not part of the inference path (Remark 7).
- *Variance collapse* (Prop. 3): the classifier consumes the deterministic h at both training and evaluation, so $\sigma^{2*} = \tau^2$ for every β (the variance decouples from the prediction term; Lemma 3), the side path keeps a finite covariance and the bound stays well-defined (Remark 10), and the trivial collapse $\mu \equiv 0$ pays the fixed positive price $\text{KL} = \frac{1}{2}a^2 > 0$ for every class — the degenerate corner is excluded by geometry rather than by the optimizer’s good behavior. This channel is closed cleanly, by construction.
- *Certificate-object mismatch* (Prop. 4): the classifier consumes the deterministic h at both training and evaluation — there is no training-code/evaluation-mean split — but the bound is an upper bound on a *noisy side-path copy* of h , not on $I(X; H|Y)$, which no finite certificate bounds. This channel is *not* closed; what FGIB buys is that the perturbation is side-path-only, and its size is governed by the conditioning of A (Prop. 12(iii)).

PART II. THE FGIB THEORY

FGIB notation. The deterministic main path computes $h = f_{\text{enc}}(X)$ and the classifier consumes h directly; the side path parameterizes $q(z|x) = \mathcal{N}(\mu(x), \Sigma(x))$ with linear mean head $\mu(x) = Ah(x)$ for $A \in \mathbb{R}^{z \times z}$ (the theory assumes $z = d$ and A invertible; the inverse transforms of Propositions 5 and 11 are defined under this assumption), and a separate logvar head for $\Sigma(x)$. The side path is *not part of inference*: at evaluation the model returns the classifier output and never evaluates the mean head (so A is a trainable module that exists only inside the regularizer — structurally the same position CEB’s backward encoder occupies on the other side of the KL). The label-conditional prior is the *fixed* anchor $r(z|y) = \mathcal{N}(av_y, \tau^2 I)$ with anchor variance $\tau^2 > 0$ (default 1.0): the v_y are orthonormal directions obtained by QR-decomposing a random matrix ($K \leq z$; the orthonormal factor of the random matrix), scaled by the anchor scale a (default 4.0), stored as non-trainable buffers. At the variance optimum of the KL the posterior variance matches

the anchor variance, $\sigma^{2*} = \tau^2$; we write σ_p^2 for this common variance where the distinction does not matter. The orthogonality of the method lies in this fixed frame and in the equilibrium it induces — the KL drives the class-conditional means $\mu_y = A \mathbb{E}[h|y]$ toward the orthonormal anchors av_y , so the transformed class representations become mutually orthogonal (the aligned family below) — while A itself remains an unconstrained linear map. For regression, fixed random-Fourier anchors $\mu_p(y) = a\sqrt{2/z} \cos(2\pi\omega y + b)$ with frozen $\omega \sim \mathcal{N}(0, 2^2)$, $b \sim U[0, 2\pi)$ replace the per-class table ($\|\mu_p(y)\| \approx a$). The loss is $L_{\text{FGIB}} = \mathbb{E}[\text{CE}(c(h), y)] + \beta \mathbb{E}[\text{KL}(q(z|x) \| r(z|y))]$, with gradients reaching the main path only through the shared encoder; the zero-initialized logvar head fixes $\sigma^2 = 1$ at initialization, which equals τ^2 in the default configuration $\tau^2 = 1$ (for general τ the theory’s variance optimum $\sigma^2 = \tau^2$ is reached by the logvar head, whose initialization is a scheduling choice), so training starts at $\text{KL} \approx \frac{1}{2\tau^2} a^2$.

Implementation caveat (theory–code layering). The theory works with a fixed orthonormal factor, an invertible linear side head $T(h) = Ah$, and per-class Gaussian priors with anchor variance τ^2 . The code implements an *affine* head (`nn.Linear` with bias) — which the theory absorbs as an idealization, the bias being immaterial when A is learned jointly — and does not enforce orthogonality, invertibility, or conditioning constraints on A . The guarantees below (orthonormal frames, invariance, conditioning) are properties of the idealized model; the implementation approximates them, and the guarantees do not transfer automatically (Remarks 7, 9). **Distributional standing assumption.** Several results state that a deterministic continuous representation carries infinite conditional information. Those statements assume the standard continuous setting: $X | Y$ is non-atomic, and the induced representation $H(X) | Y$ is a non-atomic continuous variable. For the finite empirical distribution of the training data no such conclusion is claimed; there the quantities are finite and only their growth with sample size is governed by these results.

A.1 THE EQUIVALENCE: THE SIDE PATH IS A CEB BOTTLENECK ON THE MAIN PATH

Motivation. Stochastic bottlenecks such as VIB and CEB are *stochastic at training time but deterministic at evaluation time*: the classifier is trained on sampled codes $z \sim q(z|x)$ yet evaluated on the mean $\mu_q(x)$ (Fischer 2020, Appendix A.1), so its training and evaluation distributions differ. In the same models the cross-entropy gradient and the KL gradient act on the same heads: the classification term can exploit the variance (by Jensen, $\mathbb{E}_z[\text{CE}(c(z), y)] \geq \text{CE}(c(\mu_q), y)$, so loosening the noise relaxes the upper bound), while the KL term pushes the posterior back toward the prior. Finally, the regularizer constrains the noisy code z , whereas inference consumes the noise-free mean — the object being regularized is not the object being used. FGIB resolves all three mismatches by design: the classifier consumes the deterministic representation h at both training and evaluation time, the two objectives act on disjoint parameter sets (the classification gradient does not enter the side heads, and the KL gradient reaches the main path only through the shared encoder), and, by Proposition 5 below, the side-path regularizer is, *at each fixed value of the side head*, a CEB bottleneck read in the coordinates in which the posterior is centered at the representation that inference actually uses (Remark 7 states why this is weaker than a fixed constraint on h). Confining the stochasticity to a side path also keeps the variational objective well-defined: for continuous deterministic representations the residual information diverges (Amjad & Geiger; Kolchinsky et al.), so a purely deterministic bottleneck admits no finite variational bound, while the finite side-path covariance of FGIB does (Remark 10). Whether the robustness benefits of the stochastic bottleneck survive the deterministic inference path is the empirical question studied in the experiments.

Lemma 1 (Linear invariance of the Gaussian KL). *Let $q = \mathcal{N}(\mu_q, \Sigma_q)$ and $p = \mathcal{N}(\mu_p, \Sigma_p)$ on $\mathbb{R}^z$, and let $A \in \mathbb{R}^{z \times z}$ be invertible. Define the common pushforwards $q' = A_{\#}q = \mathcal{N}(A\mu_q, A\Sigma_qA^\top)$ and $p' = A_{\#}p$. Then $\text{KL}(q' \| p') = \text{KL}(q \| p)$.*

Proof. With $d = z$, the closed form reads $\text{KL}(q \parallel p) = \frac{1}{2} [\text{tr}(\Sigma_p^{-1} \Sigma_q) + (\mu_q - \mu_p)^\top \Sigma_p^{-1} (\mu_q - \mu_p) - d + \ln |\Sigma_p| - \ln |\Sigma_q|]$. Under the common map A the three non-constant terms are unchanged:

$$\begin{aligned} \text{tr}((A \Sigma_p A^\top)^{-1} (A \Sigma_q A^\top)) &= \text{tr}(A^{-\top} \Sigma_p^{-1} A^{-1} A \Sigma_q A^\top) = \text{tr}(A^{-\top} \Sigma_p^{-1} \Sigma_q A^\top) \\ &= \text{tr}(\Sigma_p^{-1} \Sigma_q A^\top A^{-\top}) = \text{tr}(\Sigma_p^{-1} \Sigma_q), \end{aligned}$$

$$(A \Delta \mu)^\top (A \Sigma_p A^\top)^{-1} (A \Delta \mu) = \Delta \mu^\top A^\top A^{-\top} \Sigma_p^{-1} A^{-1} A \Delta \mu = \Delta \mu^\top \Sigma_p^{-1} \Delta \mu,$$

$$\ln |A \Sigma_p A^\top| - \ln |A \Sigma_q A^\top| = (\ln |\Sigma_p| + 2 \ln |A|) - (\ln |\Sigma_q| + 2 \ln |A|) = \ln |\Sigma_p| - \ln |\Sigma_q|,$$

where $\Delta \mu = \mu_q - \mu_p$ and the trace step uses the cyclic property. $\square$ $\square$

Lemma 2 (Invariance of the CEB objective and the MNI point). *For any bijective measurable map ϕ , $I(X; \phi(Z)|Y) = I(X; Z|Y)$ and $I(Y; \phi(Z)) = I(Y; Z)$. Consequently the CEB objective (Eq. 5) is invariant under bijections of Z , and Z attains the MNI point (Eq. 1) if and only if $\phi(Z)$ does.*

Proof. Mutual information is invariant under bijective transformations of either argument (data processing inequality applied in both directions). $\square$ $\square$

Proposition 5 (The FGIB side path is a CEB bottleneck on the main path, at each fixed side head). *Let $z = d$ and let the learned linear head $T(h) = Ah$ be invertible (unconstrained, so a.s. so at initialization). The fixed prior $r(z|y) = \mathcal{N}(\mu_p(y), \Sigma_p(y))$ — for FGIB, the orthonormal anchor table $\mu_p(y) = a v_y$, $\Sigma_p = \tau^2 I$ — is, by construction, the image under T of the canonical prior $r'(\tilde{z}|y) = \mathcal{N}(A^{-1} \mu_p(y), A^{-1} \Sigma_p(y) A^{-\top})$ (since $AA^{-1} = I$), and the posterior $q(z|x) = \mathcal{N}(Ah(x), \Sigma_q(x))$ is the image of $q'(\tilde{z}|x) = \mathcal{N}(h(x), A^{-1} \Sigma_q(x) A^{-\top})$. At a fixed value of A , the prior is never transformed by training: A^{-1} only reads the fixed anchor geometry in the coordinates in which the posterior is centered at h ; during training A moves, so the pulled-back prior moves with it (Remark 7). By KL invariance under the common invertible linear map $T^{-1} = A^{-1}$ (Lemma 1), pointwise for every (x, y) ,*

$$\text{KL}(q(z|x) \parallel r(z|y)) = \text{KL}(q'(\tilde{z}|x) \parallel r'(\tilde{z}|y)), \quad (10)$$

That is, at each fixed side head the side-path regularizer is pointwise identical to a CEB bottleneck with a backward encoder frozen at the current head, applied to a noisy version of the main-path representation, $\tilde{z} = h + \tilde{\epsilon}$ with $\tilde{\epsilon} \sim \mathcal{N}(0, A^{-1} \Sigma_q A^{-\top})$, using the canonical prior r' as that backward encoder. For FGIB the canonical backward encoder is the frame $\mathcal{N}(a A^{-1} v_y, \tau^2 A^{-1} A^{-\top})$, whose mean anchors $A^{-1}(a v_y)$ are orthonormal at scale a in the induced metric $A^\top A$ ($\langle A^{-1} v_i, A^{-1} v_j \rangle_{A^\top A} = \delta_{ij}$): the orthogonality enters through the fixed frame $\{v_y\}$ and the equilibrium it induces, not through a constraint on A . Hence the regularizer’s gradient with respect to h (through the shared encoder) coincides, up to the fixed linear map $A^\top$, with the gradient that a CEB model applies to its representation mean.

Proof. The map $z \mapsto \tilde{z} = A^{-1}z$ is a bijection, and $q' = (A^{-1})_\# q$, $r' = (A^{-1})_\# r$: $A^{-1}Ah = h$ and $A^{-1}\Sigma_q A^{-\top}$ give the canonical posterior, and $A^{-1}\mu_p$, $A^{-1}\Sigma_p A^{-\top}$ give the canonical prior, with $A(A^{-1}\mu_p) = \mu_p$ and $A(A^{-1}\Sigma_p A^{-\top})A^\top = \Sigma_p$, so r is exactly the image of r' under T . Equation (10) is Lemma 1 applied to A^{-1} . Sampling $\tilde{z} \sim q'$ is equivalent to $\tilde{z} = h + \tilde{\epsilon}$ with the stated noise. Combined with the variational bound of Fischer (2020, Eqs. 9–11), the regularizer upper-bounds the residual information of the noisy main-path representation: $I(X; \tilde{Z}|Y) \leq \mathbb{E}[\text{KL}(q'(\tilde{z}|x) \parallel r'(\tilde{z}|y))]$. $\square$ $\square$

Remark 7 (Trainable-implementation caveat: the side head is a second trainable party in the matching). *The identity of Proposition 5 holds pointwise at each value of the side head, so the asymmetry advertised against CEB — one trainable side versus two — is weaker than it looks. In the idealized theory A may be treated as fixed (orthonormal or any fixed invertible map); in the trainable implementation A is free and unused at inference: like CEB’s backward encoder $b(z|y)$, it lives inside the KL and can lower it without moving the deployed representation h . Since h is linear-separable by the jointly trained classifier, the linear fit of the anchor target av_y on h fits well, and much of that fit can be absorbed by A alone. The variance term $\frac{z}{2}(\sigma^2/\tau^2 - 1 - \ln(\sigma^2/\tau^2))$ does not depend on A at all, so nothing in the regularizer resists A drifting toward a projector onto the K -dimensional*

between-class subspace with a vanishing component on within-class directions. Full collapse is blocked — with invertible A , $\text{KL} = 0$ forces h itself to be class-constant — but the approach to the rank- K regime is not blocked, and that regime is exactly where the conditioning failure and the perturbation blow-up of Propositions 12 and 11 live. This is the one channel of Part I that FGIB has not closed; the fixed-orthogonal configuration of Proposition 6 below closes it by construction, and its conditioning is measured for the trainable variant in the main paper (Section 4.5); Section A.4 collects it.

Proposition 6 (Fixed orthogonal side head: the regularizer anchors the deployed representation directly). *Let the side head be a fixed orthogonal map $A \in \mathbb{R}^{d \times d}$ with $z = d$, and let $\{v_y\}$ be the anchor frame. Then:*

- (i) Isometry. $\|Ah - av_y\| = \|h - aA^\top v_y\|$, and $\{A^\top v_y\}$ is orthonormal — combined with Proposition 7, the regularizer is exactly the fixed-anchor center loss $\frac{1}{2\tau^2}\|h - aA^\top v_y\|^2$ on the deployed representation h itself, with the rotated frame $\{A^\top v_y\}$.
- (ii) Conditioning. $A^\top A = I$ has $\kappa = 1$ at every step: the pulled-back reference of Proposition 5 is orthonormal at scale a in the induced metric, the perturbation of Proposition 11 is exactly $\tau \epsilon$ with unit covariance — no blow-up — and the counterexample of Proposition 12(iii) is excluded by construction.
- (iii) Statistical equivalence. For a Haar-distributed random orthogonal A , the rotated frame $\{A^\top v_y\}$ is distributed identically to $\{v_y\}$ across seeds, so the identity map $A = I$ is a representative canonical choice: the “ $A=I$ ” configuration loses nothing over any other fixed orthogonal map.

Proof. (i) is the isometry of orthogonal maps: $\|Ah - av_y\|^2 = \|A(h - aA^\top v_y)\|^2 = \|h - aA^\top v_y\|^2$, and $v_i^\top AA^\top v_j = v_i^\top v_j = \delta_{ij}$. (ii) $\kappa(I) = 1$; the induced metric is $A^\top A = I$; the perturbation covariance is $\tau^2(A^\top A)^{-1} = \tau^2 I$. (iii) The QR frame $\{v_y\}$ is constructed from a standard normal matrix, whose distribution is invariant under the orthogonal group; Haar-distributed A maps it to another frame with the same uniform distribution on the Stiefel manifold, so the seed distribution of the whole optimization problem — encoder initialization, anchor frame, classifier initialization — is identical for $A = I$ and for any random fixed orthogonal A . $\square$ $\square$

Remark 8 (Family closure). *Lemma 1 holds for the underlying Gaussian distributions. Within a restricted variational family the equality of Proposition 5 is exact only if the family is closed under the inverse map A^{-1} : the full-covariance family is closed (and is the parameterization used by Fischer 2020); the diagonal family is not — for diagonal q, r the transformed pair is diagonal only if A^{-1} preserves diagonality (e.g., permutations and sign flips) and approximate otherwise. The diagonal implementation of the codebase is therefore covered at the distribution level, with the family approximation confined to the noise covariance $A^{-1}\Sigma_q A^{-\top}$ (Remark 19).*

Remark 9 (Rank deficiency). *If $z < d$ or A is rank-deficient, only the projection of h onto the row space of A (the image of $A^\top$ in h -space) receives gradient from the regularizer; the null-space component is unconstrained. In the default configuration $z = d$ and a randomly initialized A is almost surely invertible, but invertibility carries no conditioning margin — near-singularity is enough to produce the effects of Proposition 12(iii), and it is the direction the gap gradient points in (Remark 7). If $A \in \mathbb{R}^{z \times d}$ has full column rank ($z \geq d$), a left inverse B with $BA = I$ exists and the data-processing inequality for KL gives, per (x, y) , $\text{KL}(q\|r) \geq \text{KL}(B_\# q \| B_\# r) \geq I(X; BZ | Y)$ — a valid lower-level bound, but one whose equality (invariance) is lost, and whose pulled-back prior again depends on B . The exact invariance argument of Proposition 5 genuinely requires the invertible linear map $T(h) = Ah$ with $z = d$.*

Remark 10 (Noise-free limit). *The equivalence of Proposition 5 should be understood at the level of the variational objective, not as a mutual-information limit: as the side-path noise vanishes, $I(X; Z|Y)$ diverges for continuous deterministic H under the standing non-atomic assumption (the pathology of deterministic networks discussed by Amjad & Geiger and Kolchinsky et al.), and the variational KL bound becomes loose. Keeping the side-path covariance finite — without injecting noise into the inference path — is precisely what keeps the regularizer well-defined; this is the design rationale of FGIB.*

A.2 THE SWEEP

The degeneracy to be addressed (restated in our terms). When $Y = f(X)$, the IB curve saturates the data-processing bound and is piecewise linear, $F(r) = \min\{r, H(Y)\}$ (Caveats paper, Section 3). Two consequences (their Caveat 1): (i) by the DPI, $L_\beta^{\text{IB}}(T) = I(Y; T) - \beta I(X; T) \leq (1 - \beta)I(Y; T) \leq (1 - \beta)H(Y)$, and $T_{\text{copy}} := f(X) = Y$ attains the bound for *every* $\beta \in [0, 1]$; all $\beta \in (0, 1)$ are pinned at the single corner point $\langle H(Y), H(Y) \rangle$ of the curve; (ii) conversely, at $\beta = 1$ every point of the increasing part simultaneously maximizes L_1^{IB} . The mechanism is exposed by writing $\max_T L_\beta^{\text{IB}} = \max_r [F(r) - \beta r]$ (the Legendre transform of $-F$): on the linear increasing part every point has the *same* derivative $F'(r) = 1$, hence the same supporting slope — a Lagrangian of the form “prediction $-\beta \cdot$ compression” sweeps a Pareto curve only if each point of the curve has a *unique* supporting slope, i.e. the curve is strictly concave. The dIB variant fails identically: on the hard-clustering family $T = g(Y)$ one has $H(T|Y) = 0$, hence $I(Y; T) = H(T)$, so $L_\beta^{\text{dIB}} = (1 - \beta)H(T)$ — a *monotone* function of the single compression scalar $H(T)$, maximized by the finest partition for every $\beta < 1$. The squared fixes repair this by making the objective strictly concave in the compression scalar: $F'(r)/(2r) = \beta$ is strictly decreasing, so each β selects a unique r (Eq. 8 argument), and squared-dIB has the unique maximizer $H(T) = 1/(2\beta)$ (their Eq. A20). By Proposition 1 this degeneracy is inherited verbatim by CEB ($\beta_{\text{IB}} = 1 + \gamma > 1$), which is the first channel of Part I.

Reduced model of FGIB. The KL depends on the side mean μ alone, and the classifier term is invariant under the coordinate change $\tilde{w}_j = A^{-\top} w_j$ (since $w_j \cdot h = \tilde{w}_j \cdot \mu$): with A invertible ($z = d$ in the default configuration, so a random A is a.s. invertible), every equilibrium in (h, w) coordinates maps bijectively to one in $(\mu, \tilde{w})$ coordinates. We therefore analyze the reduced model $\mu(x) = h(x)$ without loss of generality. Consider the *class-symmetric aligned family*: the encoder maps every input of class y to $h_y = s v_y$ ($s \geq 0$), the classifier is held at the symmetric configuration $w_j = c v_j$, $b_j = 0$ with fixed scale $c > 0$, and the posterior is $q_y = \mathcal{N}(s v_y, \sigma^2 I)$. The family is the natural equilibrium manifold of the two forces: the KL gradient $\nabla_h \text{KL} = (\mu - a v_y)/\tau^2$ acts exactly along v_y , and the classification force on the class center acts along v_y at the symmetric configuration; the ansatz is a *working approximation*: it captures the direction of both forces, and is motivated as exact in the β -dominant regime, but that exactness is not proved here — the cross terms are of order $O(c/\beta)$ and are idealized away (see Remark 16).

Lemma 3 (FGIB Lagrangian on the aligned family). *On the aligned family,*

$$\text{KL}(s, \sigma^2) = \frac{z}{2}(\sigma^2/\tau^2 - 1 - \ln(\sigma^2/\tau^2)) + \frac{1}{2\tau^2}(s - a)^2, \quad \text{CE}(s) = \ln(1 + (K - 1)e^{-cs}), \quad (11)$$

and the optimizer over σ^2 is $\sigma^2 = \tau^2$ for every β , since the CE term does not depend on σ^2 (the classifier consumes the deterministic h , not z).

Proof. With $h_y = s v_y$ and $w_j = c v_j$, the logits are $c \langle v_j, s v_y \rangle = cs \delta_{jy}$, so $p_y = e^{cs}/(e^{cs} + K - 1)$ and $\text{CE} = -\ln p_y = \ln(1 + (K - 1)e^{-cs})$. For the KL, the diagonal closed form with $\Sigma_q = \sigma^2 I$, $\Sigma_p = \tau^2 I$ reads $\frac{1}{2} \sum_d [\sigma^2/\tau^2 + (\mu_d - a v_{yd})^2/\tau^2 - 1 - \ln(\sigma^2/\tau^2)]$, giving Eq. (11) since $\|v_y\| = 1$. The variance term is non-negative with a unique minimum at $\sigma^2 = \tau^2$, and CE is independent of σ^2 . $\square$

Proposition 7 (A deterministic main path freezes the variance: the released KL is exactly a fixed-variance displacement term). *Consider FGIB with a deterministic main path (the classifier consumes h , never z), a log-variance head $\ell_q(x) = Wh(x) + b$ with zero initialization, a frozen prior log-variance $\ell_p = \ln \tau^2 = 0$, and a first-order optimizer without weight decay. Then $\ell_q \equiv \ell_p$ at every training step and the regularizer is, throughout training,*

$$\text{KL}(q \| r) = \frac{1}{2\tau^2} \|\mu - \mu_p\|^2, \quad (12)$$

a fixed-variance displacement term — the KL of Lemma 3 evaluated permanently at the variance optimum $\sigma^2 = \tau^2$, not merely in the limit. More generally, when the variance enters the objective only through the KL, the per-coordinate gap $u = \ell_q - \ell_p$ obeys $\partial_u \text{KL} = \frac{1}{2}(e^u - 1)$, which carries the sign of u : $u = 0$ is the unique attractor of the continuous descent flow $\dot{u} \propto -(e^u - 1)$, and of discrete first-order updates whenever the standard step-size condition holds; zero initialization selects this fixed point at the first step and no force ever leaves it.

Proof. The per-coordinate diagonal KL $\frac{1}{2}[\ell_p - \ell_q + e^{\ell_q - \ell_p} + (\mu - \mu_p)^2 e^{-\ell_p} - 1]$ gives $\partial_{\ell_q} \text{KL} = \frac{1}{2}(e^{\ell_q - \ell_p} - 1)$, which vanishes iff $\ell_q = \ell_p$, independently of the mean gap. With $W = 0$, $b = 0$ and $\ell_p = 0$ the head output is identically zero, so $\ell_q = \ell_p = 0$ at initialization, the gradient of every head parameter is zero, and no weight decay or other variance-dependent term acts on the head: $\ell_q \equiv 0$ for all time. Substituting $\ell_q = \ell_p = 0$ into the closed form yields Eq. (12). For the general attractor statement, the gap evolves under gradient flow on ℓ_q as $\dot{u} = -\frac{\beta}{2}\|h\|^2(e^u - 1)$, whose only fixed point is $u = 0$ and whose linearization is $-\frac{\beta}{2}\|h\|^2$, so $u = 0$ is globally asymptotically stable (a one-dimensional monotone flow); a discrete first-order step multiplies the gap by $1 - \frac{\eta\beta}{2}\|h\|^2 \frac{e^u - 1}{u}$, which contracts $|u|$ whenever $\eta\beta\|h\|^2 < 4/\sup_{u'} \frac{e^{u'} - 1}{u'}$ — the standard condition. $\square$ $\square$

Corollary 1 (The FGIB family is a center-loss family; the variance race re-enters only through a trainable prior). *The same argument pins the posterior variance in every FGIB-family variant, so with $\tau^2 = 1$: FGIB and CentFGIB are exactly CE + $\frac{\beta}{2}\|\mu - av_y\|^2$ (frozen anchors), and TAFGIB is exactly the same loss with the anchor means trainable — the ablation table’s “coincidence” is an identity, not an empirical observation. DCEB is the single exception that leaves the family: its trainable prior log-variance has gradient $\partial_{\ell_p} \text{KL} = \frac{1}{2}[1 - e^{\ell_q - \ell_p} - (\mu - \mu_p)^2 e^{-\ell_p}]$, which at the zero-initialized point equals $-\frac{1}{2}(\mu - \mu_p)^2 < 0$ — the prior variance is pushed down while the means are unmatched, amplifying the mean-matching force by up to e^{10} at the clamp. This is Channel P3 re-entering through the prior side, and it is consistent with the measured behavior: DCEB is the only variant that collapses ($\beta = 10$, Section 4.5 of the main paper), while TAFGIB, whose prior variance is frozen, stays in the family and does not.*

Proposition 8 (Unique interior optimum per β at fixed classifier scale — the sweep). *At a fixed classifier scale c , for every $\beta > 0$, the FGIB Lagrangian on the aligned family,*

$$L_\beta(s) := \ln(1 + (K - 1)e^{-cs}) + \frac{\beta}{2\tau^2}(s - a)^2, \quad (13)$$

is strictly convex in s and has a unique minimizer $s^(\beta) > a$, given implicitly by*

$$\frac{\beta}{\tau^2}(s - a) = c\varphi(cs), \quad \varphi(u) := \frac{(K - 1)e^{-u}}{1 + (K - 1)e^{-u}} \in (0, 1). \quad (14)$$

The map $\beta \mapsto s^(\beta)$ is continuous and strictly decreasing, with $s^*(\beta) \rightarrow a$ as $\beta \rightarrow \infty$ and $s^*(\beta) \rightarrow \infty$ as $\beta \rightarrow 0$. Consequently the map $\beta \mapsto (\text{KL}(s^*), \text{CE}(s^*))$ is one-to-one onto the entire curve $\{(\frac{1}{2\tau^2}(s - a)^2, \ln(1 + (K - 1)e^{-cs})) : s \in (a, \infty)\}$: no two β share the same optimum, and every interior point of the trade-off curve is attained — the exact failure mode of L_β^{IB} (all β pinned at T_{copy}) is absent. The fixed- c condition is load-bearing and is analyzed in Remark 12.*

Proof. Compute $\text{CE}'(s) = -c\varphi(cs)$ and $\text{CE}''(s) = c^2(K - 1)e^{-cs}/(1 + (K - 1)e^{-cs})^2 > 0$, so CE is strictly convex and strictly decreasing in s ; the compression term $\frac{1}{2\tau^2}(s - a)^2$ is strictly convex. Hence $L_\beta''(s) = \text{CE}''(s) + \beta/\tau^2 > 0$: L_β is strictly convex with at most one minimizer, characterized by $0 = L_\beta'(s) = -c\varphi(cs) + \beta(s - a)/\tau^2$, i.e. Eq. (14). The left side $\beta(s - a)/\tau^2$ is strictly increasing from $-\beta a/\tau^2$ (at $s = 0$) through 0 (at $s = a$) to $+\infty$; the right side $c\varphi(cs)$ is strictly decreasing in s with $c\varphi(cs) \in (0, c)$. The unique crossing therefore satisfies $s^* > a$. Differentiating Eq. (14) with respect to β gives $(s - a)/\tau^2 + (\beta/\tau^2)s' = c^2\varphi'(cs)s'$ with $\varphi' < 0$, so $s' = -(s - a)/(\beta - c^2\tau^2\varphi'(cs)) < 0$: s^* is strictly decreasing. The limits follow by monotonicity: as $\beta \rightarrow \infty$ the left side can only match the bounded right side when $s \rightarrow a$; as $\beta \rightarrow 0$ the crossing requires $\varphi(cs) \rightarrow 0$, i.e. $s \rightarrow \infty$. $\square$ $\square$

Proposition 9 (The trade-off curve is strictly concave in the compression–prediction plane). *On the curve of Proposition 8 (fixed classifier scale c),*

$$\frac{d \text{CE}}{d \text{KL}} = \frac{\text{CE}'(s)}{(s - a)/\tau^2} = -\frac{\tau^2 c\varphi(cs)}{s - a}, \quad (15)$$

which is strictly increasing from $-\infty$ (as $s \rightarrow a^+$) to 0^- (as $s \rightarrow \infty$). Thus the curve $\text{KL} \mapsto -\text{CE}$ (compression versus achievable prediction, on the KL–CE surrogate plane of Remark 11) is strictly concave, and each of its points has a unique supporting slope $-\beta$. By the criterion of the Caveats paper (their Fig. 2 right panel and the $F'(r)/(2r)$ argument of Eq. 8), this is precisely the property

that makes the Lagrangian $-\text{CE} - \beta \text{KL}$ sweep the full surrogate curve on the aligned family at fixed scale: the convexity of the compression term supplies the strict-concavity condition without squaring it. This is a statement about the KL-CE surrogate of Proposition 8, not about the Caveats information plane (Remark 11).

Proof. φ is positive and strictly decreasing and $s - a$ is positive and strictly increasing, so $\tau^2 c \varphi(cs)/(s - a)$ is strictly decreasing in s ; the signs give the stated limits. The stationarity condition Eq. (14) is exactly $-d\text{CE}/d\text{KL} = \beta$, i.e. β is the supporting slope at the optimum. $\square \square$

Remark 11 (What the sweep sweeps: a KL-CE surrogate curve, not the IB curve). Propositions 8 and 9 prove strict convexity of the FGIB Lagrangian and one-to-one parameterization by β on the class-symmetric aligned family, at fixed classifier scale, in the plane $(\text{KL}_{\text{anchor}}, \text{CE})$. That plane is not the Caveats paper’s IB plane $(I(X; T), I(Y; T))$, and the claim does not extend to it. On the aligned family the posterior is class-constant by construction, so $I(X; Z|Y) = 0$ for every s , while the deterministic class prototypes $h_y = sv_y$ already retain the label completely for any $s > 0$ ($I(Y; H) = H(Y)$); what varies along the sweep is the anchor mismatch and the classifier margin, not a proved compression of mutual information. Whether the deployed representation’s residual $I(X; H|Y)$ moves monotonically with β is a separate, empirical question; the theory guarantees a strictly concave surrogate, not a recovered IB curve. This qualification is carried into Section A.4.

Remark 12 (The classifier scale is load-bearing, and the current implementation leaves it free). If c is free, the pair $(s, c) = (a, c)$ with $c \rightarrow \infty$ sends $\text{KL} = \frac{1}{2\tau^2}(s - a)^2 = 0$ and $\text{CE} = \ln(1 + (K - 1)e^{-ca}) \rightarrow 0$ simultaneously (for $a > 0$): the infimum of L_β over the aligned family is 0 for every β , no minimizer is attained, and the one-to-one sweep of Proposition 8 has no global counterpart. (At $a = 0$ the margin growth alone cannot drive CE to zero.) This is the FGIB analogue of the dead knob — the same one-corner attractor, now arrived at by growing the margin rather than by the linearity of the deterministic IB curve. Three consequences. (a) The present implementation is exposed. The classifier is `nn.Linear` trained by Adam with no weight decay, so c is unconstrained. On separable data the CE gradient drives $\|w_y\|$ upward; it does not diverge in finite time (the gradient vanishes as $p_y \rightarrow 1$, giving logistic growth), but at any finite scale c the equilibrium $s^*(\beta; c)$ sits on the curve of Proposition 8 for that c , and s^* drifts toward a as c grows. (b) Testable prediction. The β -dependence of the equilibrium should decay with training length; since the experiments use early stopping with `patience = 10`, part of any observed β -effect is plausibly an early-stopping effect. This can be tested by training well past saturation and comparing the converged KL across β . (c) Architectural fix. The device that satisfies the fixed- c assumption by construction is a cosine classifier: normalized features, normalized rows, and a fixed temperature (scale). Weight normalization with a learnable gain does not guarantee a bounded scale, and weight decay only replaces the free-scale problem with a joint optimization problem over (s, c) that must be re-derived — it mitigates the divergence but does not make Proposition 8 exact. Under a fixed-norm classifier with fixed temperature the proposition holds as stated, at the price of one architecture change.

Where the convexity comes from (mechanism). Compare the three objectives on their natural solution families, as functions of the single compression scalar:

- *dIB on hard clusterings* (Caveats Appendix B): $L_\beta^{\text{dIB}} = (1 - \beta)H(T)$ — compression $H(T)$ and prediction $I(Y; T) = H(T)$ are the same scalar, so the objective is monotone in it and the optimum sits at the endpoint for every $\beta < 1$. Squared-dIB restores interior optima by hand, $H(T) - \beta H(T)^2$ with unique maximizer $H(T) = 1/(2\beta)$.
- *Standard IB* (deterministic): the curve is linear ($F'(r) \equiv 1$), every point shares one supporting slope, and the corner $\langle H(Y), H(Y) \rangle$ is selected for all $\beta \in (0, 1)$. Squared-IB convexifies via $F'(r)/(2r)$ strictly decreasing.
- *FGIB*: the compression term is $\text{KL}(q_y \| r_y) = \frac{1}{2\tau^2} \|\mu_y - av_y\|^2 + (\text{variance terms})$, equal to $\frac{1}{2\tau^2}(s - a)^2$ at the variance optimum — quadratic in the alignment scalar s , hence strictly convex, because the KL is measured against a fixed reference av_y : $\text{KL}(\mathcal{N}(\mu, \cdot) \| \mathcal{N}(\mu_p, \cdot)) = \frac{1}{2\tau^2} \|\mu - \mu_p\|^2 + (\text{variance terms})$ is quadratic in the displacement $\mu - \mu_p$ whenever μ_p does not move. Entropy-based compression ($H(T)$, $I(X; T)$) is self-referential — it depends only on the representation’s own distribution and enters linearly

in the clustering probabilities; the fixed anchor supplies an absolute reference frame and turns compression into a squared distance. The convexification that squared-IB installs by squaring the compression term is thus built into FGIB’s fixed-anchor KL. Together with the strictly convex prediction term (Lemma 3), the Lagrangian is strictly convex with a unique interior minimizer per β at fixed classifier scale (Proposition 8), and the resulting curve is strictly concave (Proposition 9). With the scale free, the same quadratic gap is driven to zero by margin growth alone (Remark 12) — the convexification is necessary but not sufficient without the scale condition.

Remark 13 (The compressed endpoint is non-collapsed but still a label code — Caveat 2 remains). *VIB’s fixed marginal $m(z) = \mathcal{N}(0, I)$ places the KL = 0 point at $\mu = 0$ for all classes: the fully compressed solution is the trivial collapse with CE = $\ln K$ (chance level), and squaring (SVIB) leaves this corner in place since KL^2 is stationary at $\text{KL} = 0$. FGIB’s per-class orthonormal anchors make the KL = 0 point a class-separated configuration: $\mu_y = av_y$ with $\langle v_y, v_{y'} \rangle = \delta_{yy'}$, so at the corner $\text{CE}(a) = \ln(1 + (K - 1)e^{-ca})$, which is bounded and tends to 0 as the anchor scale a grows, and $a = 0$ recovers the VIB corner. But class-separation does not buy non-triviality in the sense of the Caveats paper: their Caveat 2 identifies as uninteresting the variables on the manifold T_α that mix a constant map with an exact copy of Y , compressing by “forgetting the input with some probability” rather than by merging perceptually similar classes — and FGIB’s compressed corner $\mu_y = av_y$ is a deterministic function of Y alone, i.e. a label code of exactly that kind. The orthonormal frame is in fact opposed to the kind of compression Caveat 2 holds up as interesting: it places every pair of classes at the same distance and can never express “merge collie with retriever, keep both far from teapot”. What the anchors do is choose one specific label-code geometry — class means on a scaled simplex, a neural-collapse-like configuration — instead of leaving it to the optimizer, and refuse the chance-level corner. The compressed endpoint is thus designed, not certified, and any claim that the method compresses input semantics at that endpoint is exactly the claim Caveat 2 rules out.*

Remark 14 (Residual-information interpretation (CEB)). *For any backward encoder, $\mathbb{E}[\text{KL}(q(z|x) \| r(z|y))] = I(X; Z|Y) + \mathbb{E}_y[\text{KL}(q(z|y) \| r(z|y))] \geq I(X; Z|Y)$ (Fischer 2020, Eqs. 9–11, 20; Lemma 4). FGIB is therefore a CEB bottleneck with a fixed backward encoder: the regularizer upper-bounds the residual information, and on the aligned family the residual vanishes identically ($q(z|x)$ depends on y alone), so the regularizer equals the anchor-matching gap $\frac{1}{2\tau^2}(s - a)^2$. The quadratic mechanism of Proposition 8 is inherited from this gap term, which exists only because the anchor is not allowed to follow the encoder.*

Remark 15 (Variance decoupling and training–evaluation consistency). *Since the classifier consumes the deterministic h (and the KL does not see the classifier), the posterior variance decouples: $\sigma^{2*} = \tau^2$ for all β , and the trade-off is purely geometric (mean alignment). This contrasts with VIB, where the classifier consumes the sampled z : the noise couples into the prediction term (by Jensen, $\mathbb{E}_z[\text{CE}(c(z), y)] \geq \text{CE}(c(\mu), y)$), the rate $R \geq I(X; Z)$ over-counts compression, and training-time sampling versus evaluation-time mean creates a distribution mismatch (Fischer 2020, Appendix A.1).*

Remark 16 (Idealizations). *The class-symmetric ansatz idealizes the equilibrium geometry; it is exact in the β -dominant regime and captures the direction of the KL force in general. For regression, the continuous RFF anchors satisfy $\|\mu_p(y)\| \approx a$, so the KL again contains the quadratic displacement term $\frac{1}{2}\|\mu - \mu_p(y)\|^2$ + (variance terms) — the same convexity mechanism carries over, again subject to the fixed-scale condition of Remark 12 (the regression classifier has no margin scale; its norm is likewise unconstrained in the implementation). Training starts at $\text{KL} \approx a^2/2$ (zero-initialized logvar head, $\sigma = 1$, $\mu \approx 0$), so the anchor scale sets the initial regularization pressure; larger a delays the approach to s^* . In the released implementation the variance does not merely start there: by Proposition 7 it is pinned there for the whole run, and $\text{KL} = \frac{1}{2\tau^2}\|\mu - \mu_p\|^2$ exactly. The finite-scale caveat formerly in this remark is stated, with its fix, in Remark 12.*

A.3 THE CERTIFICATE

Lemma 4 (Gap identity). *For any conditional $r(z|y)$,*

$$\mathbb{E}[\text{KL}(q(z|x) \| r(z|y))] = I(X; Z|Y) + \mathbb{E}_y[\text{KL}(q(z|y) \| r(z|y))]. \quad (16)$$

Proof. $\mathbb{E}[\text{KL}(q(z|x)||r(z|y))] = \mathbb{E}[\ln q(z|x) - \ln r(z|y)]$ and $I(X; Z|Y) = \mathbb{E}[\ln q(z|x) - \ln q(z|y)]$ (Fischer 2020, Eqs. 9–10), where both expectations are over $p(x, y)q(z|x)$; subtracting, $\mathbb{E}[\ln q(z|y) - \ln r(z|y)] = \mathbb{E}_y[\text{KL}(q(z|y)||r(z|y))]$ since the argument depends on z, y only. $\square$

Lemma 5 (Moment matching). *Within the per-class Gaussian family with diagonal covariance, the gap of Lemma 4 is minimized by the moment-matched Gaussian $b_y^* = \mathcal{N}(m_y, S_y)$ with $m_y = \mathbb{E}q(z|y)$, $S_y = \text{diag Var } q(z|y)$.*

Proof. $\text{KL}(p||q)$ as a function of q is minimized by matching the first two moments of p (standard; for Gaussians the minimizer is unique). $\square$

Proposition 10 (Variational ordering: CEB’s bound is tighter in value). *For every fixed encoder $q(z|x)$, $\text{ReX}(\theta^*) \leq \text{ReX}_F$ where θ^* is CEB’s optimal backward encoder, and the excess is the anchor gap*

$$\Delta := \text{ReX}_F - \text{ReX}(\theta^*) = \mathbb{E}_y[\text{KL}(q(z|y) \| N(av_y, \tau^2 I))] - \mathbb{E}_y[\text{KL}(q(z|y) \| N(m_y, S_y))] \geq 0, \quad (17)$$

with equality iff the anchors equal the moment-matched marginals. Here θ^ ranges over the ideal CEB backward encoder family — per-class Gaussians with free mean and free diagonal covariance (the family of Lemma 5, matching the diagonal implementation), i.e. any conditional $r(z|y)$ of that family the theory can express; the code’s single affine `prior_net` is only an approximation of this family, and the ordering is a statement about the idealized CEB, not about the implemented parameterization. On the aligned family of the sweep analysis ($q(z|y) = \mathcal{N}(sv_y, \sigma^2 I)$, so $I(X; Z|Y) = 0$ and $b^* = q(z|y)$ exactly),*

$$\Delta = \frac{1}{2\tau^2}(s - a)^2 + \frac{z}{2}(\sigma^2/\tau^2 - 1 - \ln(\sigma^2/\tau^2)), \quad (18)$$

i.e. $\text{ReX}_F = \Delta$ and $\text{ReX}(\theta^) = 0$: FGIB’s bound value equals the sweep-driving compression term of Proposition 8, while CEB’s bound collapses to the (zero) residual.*

Proof. The anchor $N(av_y, \tau^2 I)$ is a member of CEB’s per-class Gaussian family, so the minimum over the family is no larger than the value at the anchor: the ordering and Eq. (17) follow from Lemmas 4 and 5. On the aligned family the marginal is itself Gaussian, so $b^* = q(z|y)$ gives gap 0, $I(X; Z|Y) = 0$ (the posterior is class-constant), and $\text{KL}(\mathcal{N}(sv_y, \sigma^2 I) || \mathcal{N}(av_y, \tau^2 I)) = \frac{1}{2\tau^2}(s - a)^2 + \frac{z}{2}(\sigma^2/\tau^2 - 1 - \ln(\sigma^2/\tau^2))$. $\square$

Remark 17 (Tightness–force trade-off). *Proposition 10 is the price of the fixed anchor, and it is paid for a reason: in CEB the gap is a function of the backward encoder alone — $b_\theta \rightarrow q(z|y)$ eliminates it with the encoder held fixed, so CEB’s bound can become tight while the representation stays uncompressed: its tightness certifies nothing about compression, and the gap is an unobservable optimization residual that tracks the forward encoder. FGIB’s gap can shrink only if the posterior moves toward the anchors (the anchor itself cannot move), so the force is persistent and, on the aligned family, the bound value is the anchor-matching gap $\frac{1}{2\tau^2}(s - a)^2$, the sweep-driving compression term (Eq. (18)). The qualifier is necessary: “the posterior” includes the side head A , which is trainable and unused at inference, so part of the movement can be absorbed without the deployed representation h moving (Remark 7). On the aligned family, where A is fixed by the ansatz, the equivalence is exact; off it, bound tightness is monotonically equivalent to alignment of the posterior, not to compression of the deployed h .*

Proposition 11 (The inverse linear transform: the object of the bound). *Let the side head $T(h) = Ah$ be invertible ($z = d$, unconstrained), and let the trained posterior be isotropic, $\Sigma(x) = \sigma_p^2 I$ with $\sigma_p^2 = \tau^2$ the anchor variance (the KL variance term is minimized at $\sigma^2 = \tau^2$, so this holds at the variance optimum). Then $\tilde{z} := A^{-1}z$ gives $q'(\tilde{z}|x) = \mathcal{N}(h(x), \tau^2(A^\top A)^{-1})$ — the posterior is centered at the inference representation h with side noise $\tau^2(A^\top A)^{-1}$ — and the transformed anchors $r'(\tilde{z}|y) = \mathcal{N}(a A^{-1}v_y, \tau^2(A^\top A)^{-1})$ form an orthonormal frame at scale a in the induced metric $A^\top A$ ($\langle A^{-1}v_i, A^{-1}v_j \rangle_{A^\top A} = \delta_{ij}$: the fixed frame $\{v_y\}$ read in canonical coordinates). By KL invariance under invertible linear maps (Lemma 1),*

$$\text{ReX}_F = I(X; h + \tau(A^\top A)^{-1/2}\varepsilon | Y) + \mathbb{E}_y[\text{KL}(q'(\tilde{z}|y) || r'(\tilde{z}|y))], \quad \varepsilon \sim \mathcal{N}(0, I). \quad (19)$$

The bound's object is the inference representation perturbed by known, side-path-only noise (its covariance is a model parameter), and the anchor geometry is preserved exactly under the transform, in the induced metric. The perturbation, however, is small only if the side head is well-conditioned: its magnitude is governed by $A^\top A$, which is unconstrained (Proposition 12(iii)).

Proof. $A^{-1}A = I$ and $A^{-1}z = h + A^{-1}\varepsilon$ for $\varepsilon \sim \mathcal{N}(0, \Sigma)$; $A^{-1}\Sigma A^{-\top} = \tau^2 A^{-1}A^{-\top} = \tau^2(A^\top A)^{-1}$ for $\Sigma = \tau^2 I$; the prior covariance maps to $A^{-1}(\tau^2 I)A^{-\top} = \tau^2(A^\top A)^{-1}$; and $\langle A^{-1}v_i, A^{-1}v_j \rangle_{A^\top A} = v_i^\top A^{-\top} A^\top A A^{-1}v_j = \delta_{ij}$. Equation (19) is Lemma 4 applied to the transformed pair. $\square$

Proposition 12 (Orthogonal-factor conditioning: the reference is fixed and well-conditioned). *FGIB's anchor frame $V = [v_1, \dots, v_K]$ is the orthonormal factor of a random matrix (QR : $G = QR$, $V := Q$, $V^\top V = I_K$), and the anchor covariance is $\tau^2 I$ with $\tau^2 > 0$. Then, for every anchor scale $a \geq 0$ and throughout training:*

- (i) *the reference of the matching — the unscaled mean frame V and the covariance $\tau^2 I$ — is fixed (non-trainable buffers) and has condition number 1: the certificate is a distance to a compact, non-degenerate, class-separated (for $a > 0$) target. (At $a = 0$ the mean matrix aV is zero and has no condition number; the condition-1 statement attaches to V and $\tau^2 I$, not to aV .)*
- (ii) *in side-mean coordinates the certificate's displacement metric is a scaled identity: the KL's displacement term is $\frac{1}{2\tau^2} \|\mu - av_y\|^2$ with gradient $(\mu - av_y)/\tau^2$;*
- (iii) *on the shared-encoder side the metric is $A^\top A$, where A is a learnable unconstrained map, so its global condition number is not controlled by construction — and the aligned family does not control it either. An earlier version inferred condition number 1 on the class-mean subspace from the identity $h_i^\top A^\top A h_j = s^2 \delta_{ij}$; that inference is invalid, since $\{h_i\}$ is not a Euclidean orthonormal basis. Counterexample: with $A = \text{diag}(\varepsilon, 1)$, $h_1 = (1/\varepsilon, 0)$, $h_2 = (0, 1)$ and $s = 1$, one has $Ah_i = v_i$ yet $\kappa(A^\top A) = \varepsilon^{-2}$ — and the inverse-transform perturbation of Proposition 11 is correspondingly ε^{-2} -large on the first direction. The counterexample is not adversarial: it is the direction the gap gradient points in, since the KL is minimized by fitting the anchor target on h in least squares, which suppresses within-class directions (Remark 7). Any guarantee on this side therefore requires an explicit constraint on A (orthogonal parameterization, weight norm, or the like) or an empirical measurement of $\kappa(A^\top A)$ over training.*

CEB's backward encoder $b_\theta(z|y) = \mathcal{N}(m_\theta(y), S_\theta(y))$ is an unconstrained linear map of the label: both sides of its matching are trainable, its covariance diagonal can lie anywhere in the clamp $[e^{-10}, e^{10}]$ (condition number up to e^{20}), its means can be rank-deficient (class-merged codes, or the collapse solution $m_\theta \equiv 0$), and the variance race drives S_θ to the clamp, degenerating the certificate into a mean-matching force of weight $1/s = e^{10}$ (Prop. 3, subject to Remark 4). What the fixed QR reference guarantees for FGIB is therefore the reference side only: a compact, non-degenerate, class-separated target whose conditioning cannot degrade. On the encoder side FGIB's bound is exposed to the same degeneration mechanism CEB's is, with A playing the role of the second trainable party.

Proof. $V^\top V = I_K$: all singular values of V are 1; the covariance $\tau^2 I$ has all eigenvalues τ^2 ; both are frozen buffers, so (i) holds. The diagonal closed form of $\text{KL}(\mathcal{N}(\mu, \sigma^2 I) \parallel \mathcal{N}(av_y, \tau^2 I))$ has displacement term $\frac{1}{2\tau^2} \|\mu - av_y\|^2$, giving (ii). For (iii): the identity $h_i^\top A^\top A h_j = \mu_i^\top \mu_j = s^2 v_i^\top v_j = s^2 \delta_{ij}$ holds on the aligned family and only says that $\{h_i\}$ is $A^\top A$ -orthogonal by construction; it implies nothing about the spectrum of $A^\top A$, and the displayed example verifies the failure. The KL gradient on A is $(\mu - av_y) h^\top / \tau^2$, an alignment force toward the orthonormal frame that vanishes only at perfect alignment on non-degenerate samples (it also vanishes pointwise at $h = 0$, or by cancellation across a batch) — but nothing in that force resists a vanishing within-class component of A . For CEB, the displacement term is $\frac{1}{2}(m - g)^\top S_\theta^{-1}(m - g)$: the metric S_θ^{-1} has condition number up to e^{20} under the logvar clamp, and the gradient weight $1/s$ at the collapse equilibrium is Prop. 3. $\square$

Remark 18 (Two senses of tightness). *In the pure variational value sense (same encoder, same Gaussian family), CEB’s bound is tighter: $\text{ReX}(\theta^*) \leq \text{ReX}_F$ (Prop. 10), the excess being the anchor gap — the price of the fixed anchor. The object sense does not reverse this ordering: both bounds are valid upper bounds on their own noisy codes, and neither bounds the residual of a deployed deterministic continuous representation, which is infinite under the standing non-atomic assumption (Prop. 4). The structural difference is where the certified-versus- deployed gap lives — in CEB it is driven by the prediction term inside the inference path; in FGIB it is a side-path quantity whose size is governed by the conditioning of A (Prop. 12(iii)) — not which bound is “tighter” or “valid”. The learnability of the side head A bears on that difference: it is a second trainable party in the matching, unused at inference (Remark 7), so it can absorb part of the anchor gap without moving the deployed representation. On the aligned family, where A is fixed by the ansatz, its optimization drives the gap to the equilibrium value $\frac{1}{2}(s^* - a)^2$ at fixed classifier scale (Prop. 8); off the ansatz this is an open question (Section A.4).*

Remark 19 (Unconstrained side head (the implementation)). *Proposition 11 holds for any invertible linear head: the KL invariance is exact, and the object claim needs only that the perturbation covariance $\tau^2(A^\top A)^{-1}$ be known (A is a model parameter) — but the object is informative about h only insofar as that covariance is moderate, which the construction does not guarantee (Prop. 12(iii)). The diagonal variational family is not closed under A^{-1} (Remark 8), so within the diagonal parameterization of the codebase the equivalence is exact at the distribution level and approximate as a variational identity; a full-covariance side head would make it exact within the family. No orthogonality constraint on A is required — the orthogonality of the method lives in the fixed anchor frame and in the equilibrium it induces (the aligned family).*

Remark 20 (Anchor-frame reading). *With the orthonormal anchor matrix V (columns v_y , $K \leq z$), left multiplication by $V^\top$ expresses the anchor-space components of z in class coordinates: the anchors become a_{e_y} and the gap of Lemma 4 decomposes into K per-class terms plus, when $K < z$, an orthogonal-complement term that the reading ignores ($V^\top$ is a projection, not a coordinate change). The orthogonal frame is the geometric content of the prior, and it is preserved by the invertible linear pull-back A^{-1} of Proposition 11 — not by $A^\top$, which is the Jacobian factor that appears in gradients and equals A^{-1} only when A is orthogonal. An earlier version conflated the two.*

Remark 21 (Positioning against VIB and NIB). *VIB’s rate bounds the full mutual information $I(X; Z)$ with a marginal prior; CEB removes the retained label information ($\text{ReX} = R - I(Y; Z)$ at the optima), and FGIB inherits the residual-level object while replacing the learnable backward encoder with a fixed geometric prior (Prop. 10). NIB’s non-parametric bound $\hat{I} \geq I(X; M)$ is y -unconditioned and tight only when the components share a covariance and form well-separated clusters (Kolchinsky et al. 2017, Eq. 10 and the discussion following Eq. 11); when the noise is small relative to the cluster spread it saturates at $\ln N$ and its gradient vanishes (their Section IVA) — loose exactly in the regime where the parametric FGIB bound is well-behaved. The two bounds are complementary: NIB certifies the full compression non-parametrically when the geometry is favorable; FGIB bounds the residual of a noisy copy of the deployed representation parametrically, with a geometry of its own choosing — subject to the conditioning caveat of Proposition 12(iii).*

Remark 22 (Degenerate anchors). *At $a = 0$ the anchors collapse to $N(0, \tau^2 I)$ for all classes and the gap loses its class structure: $\Delta = \mathbb{E}_y[\text{KL}(q(z|y)||N(0, \tau^2 I))]$ — a VIB-style marginal gap; the structural separation of Proposition 4(b) still holds, but the compressed endpoint becomes the trivial collapse (Remark 13).*

A.4 WHAT IS PROVED AND WHAT IS OPEN

Remark 23 (Scope of the results). **Proved.** (1) The dead-knob proposition for CEB and the MNI reading of it (Prop. 1, Remark 1); (2) the gap identity, Gaussian moment matching, and the value ordering of CEB’s bound over FGIB’s at a fixed encoder (Lemmas 4, 5; Prop. 10); (3) the blind certificate on the T_α manifold and the $1:\beta$ exchange rate (Prop. 2); (4) KL invariance under invertible linear maps and the pointwise equivalence of the side path to a CEB bottleneck at a fixed side head (Lemmas 1, 2; Prop. 5); (5) the variance decoupling $\sigma^{2*} = \tau^2$ and the non-collapsed compressed endpoint (Lemma 3, with $\sigma^{2*} = \tau^2$; Remark 13); (6) the one-dimensional sweep and its strict concavity at fixed classifier scale, as a KL–CE surrogate curve on the aligned family (Props. 8, 9; Remark 11 — not a recovered IB curve); (7) the well-conditioning of the fixed reference side

(Prop. 12(i)–(ii)); (8) the deterministic main path freezes the variance: in the released implementation the regularizer is exactly the fixed-variance displacement term $\frac{1}{2\tau^2} \|\mu - \mu_p\|^2$ at every step, and the FGIB family is one center-loss family — the ablation table’s number-coincidence is an identity, not an empirical observation (Prop. 7, Cor. 1); (9) the fixed orthogonal side head anchors the deployed representation directly: $\|Ah - av_y\| = \|h - aA^\top v_y\|$ with $\kappa(A^\top A) = 1$ and unit-covariance pull-back noise, so the conditioning counterexample is excluded by construction and $A = I$ is a representative canonical choice (Prop. 6).

Measured. (i) The trainable side head A degrades $\kappa(A^\top A)$ from $10^{3.9}$ at initialization to $10^{5.8}$ – $10^{6.8}$ (transient peaks to $10^{8.3}$) on MNIST/MLP, without touching accuracy (main paper, Section 4.5, Fig. 2) — the channel of Remark 7 is real. The fixed-orthogonal variant of Prop. 6 closes it by construction, at a measured price: across five settings FGIB-H ($A=I$) trails the trainable head by ≈ 0.5 points per setting at the inherited anchor scale $a = 16$ (mean rank 4.00 of 5 vs. FGIB’s 1.20), recovering to mean rank 2.25 when the anchor scale is re-tuned for h-space ($a = 4$), so the trade-off is accuracy-versus-certification, small in both directions, and both configurations ship (main paper, Table 8). (ii) CEB’s variance race is invisible at the protocol horizon but drifts into view by 400 epochs, in the predicted direction and far from the clamp (Remark 4).

Open or conditional. (1) For the optional trainable- A variant, the bound’s object may be far from the deployed representation exactly where the residual lives (Remark 7, Prop. 12(iii)); the fixed-orthogonal configuration removes the issue by construction, and the measured question left open is only whether trainable A ever helps — on MNIST it does not. (2) The global sweep requires a bounded classifier scale (Remark 12); the current implementation leaves the scale free and the proposition conditional. A fixed-temperature cosine classifier would satisfy the fixed- c assumption by construction; weight decay only mitigates the scale divergence and replaces the problem with a new joint optimization over (s, c) that must be re-derived; the enforced-scale variants are tested on MNIST (Section 4.5 of the main paper). (3) Neither CEB’s nor FGIB’s certificate upper-bounds the deployed deterministic residual (Prop. 4); claims of inference-representation compression are hypotheses, not conclusions. (4) Caveat 2 is not resolved (Remark 13): the compressed endpoint is a chosen label-code geometry, not certified semantic compression. (5) CEB’s two-stage variance dynamics (Prop. 3(ii)–(iii)) remain a hypothesis for horizons beyond the tested 400 epochs and for protocols other than ours; the FGIB-family counterpart is proved (Prop. 7), with DCEB the single variant that re-opens the race through its trainable prior variance (Cor. 1). (6) The regression RFF anchor analysis inherits the same fixed-scale caveats and is presently a transfer argument, not a proof; the subject-disjoint AgeDB control (main paper, Table 9) is evidence, not certification.

B FAILURE CHANNELS IN DETAIL

This section states the four channels of Section 2 with the precision the main text omits; the propositions themselves are proved in Appendix A (Part I).

P1 — the knob is dead in the deterministic scenario. By the chain rule, CEB’s objective is $I(X; Z) - (1 + \gamma)I(Y; Z)$, and the data-processing inequality gives $L_\gamma(Z) \geq -\gamma I(Y; Z) \geq -\gamma H(Y)$ with equality iff $I(X; Z) = I(Y; Z)$ and $I(Y; Z) = H(Y)$. Every $\gamma > 0$ therefore has the same exact optimum — the corner — so $\gamma \mapsto (I(X; Z^*|Y), I(Y; Z^*))$ is constant and no interior point of the IB curve is ever a Lagrangian minimizer (Proposition 1). Since $\beta_{IB} = 1 + \gamma > 1$ always, CEB sits in the regime that Kolchinsky et al. (2019a) show to be degenerate over its entire parameter range. Kolchinsky et al. (2019a) themselves note that if one’s goal is the corner, this reads as β -robustness rather than failure. Remark 1 explains why that does not close the matter: the corner is one point in the information plane but an entire equivalence class of representations, and adversarial robustness, calibration and OoD detection — the properties CEB is argued for — are properties of the geometry of Z , on which the objective is exactly constant. Which member of the class is reached is settled by the optimizer, i.e. by channels P3 and P4.

P2 — a certificate that is blind, and a $1:\beta$ exchange rate. On the manifold $T_\alpha = B_\alpha \cdot Y$ of Kolchinsky et al. (2019a), T_α is a function of Y alone, so $I(X; T_\alpha|Y) = 0$ and, at the optimal backward encoder, $\text{ReX}(T_\alpha) = 0$ for every α — from total collapse ($\alpha = 0$, no label information) to full copy ($\alpha = 1$). Reading ReX as a meter of distance to the MNI point is therefore valid only along the $I(X; Z|Y)$ axis; on the necessity axis the meter reads 0 at chance level. With the optimal

classifier the objective is $\delta + \beta \text{ReX}$ where $\delta = H(Y) - I(Y; T_\alpha)$, so it is indifferent between giving up δ nats of label information and saving δ/β nats of certified residual: in the heavy-regularization regime the whole label axis is purchasable for $H(Y)/\beta$ nats of certified compression (Proposition 2).

P3 — the variance race. For the diagonal Gaussian family the KL is stationary in the posterior variance at $\sigma^2 = \tau^2$, with residual value $\|m - g\|^2/(2\tau^2)$ — pure mean matching. The cross-entropy term, however, is convex in z for a linear classifier (the log-sum-exp of an affine function of z), so by Jensen $\mathbb{E}_z[\text{CE}(c(z), y)] \geq \text{CE}(c(m), y)$ and it strictly prefers $\sigma^2 \rightarrow 0$. Along the matched line the KL resists shrinking τ^2 only while the means are unmatched ($\partial_{\tau^2} \text{KL} = -\|m - g\|^2/(2\tau^4) < 0$); once $\|m - g\| \rightarrow 0$ the race is unopposed and runs to the implementation clamp $s = e^{-10}$, at which the certificate has become a mean-matching force of weight $1/s \approx 2.2 \times 10^4$ between two trainable maps (Proposition 3). Parts (ii)–(iii) of that proposition are a direction-of-force argument, not a convergence proof. We ran the falsification test it invites, by logging the trained log-variances (Section 4.5): at the early-stopping time scale of our protocol the race does not materialize — CEB’s variances remain at their initialization $\sigma^2 = \tau^2$ with zero clamp fraction on every β — so the convergence prediction of part (iii) is empirically void under this protocol, exactly as Remark 4 allows.

P4 — the certified object is not the deployed object. At the variance optimum the posterior standard deviation is class-constant, so $Z = \mu(X) + \sqrt{S(Y)}\varepsilon$ is a noisy function of $\mu(X)$ given Y and the data-processing inequality gives $I(X; Z|Y) \leq I(X; \mu(X)|Y)$. The gap is not a small correction: whenever μ is continuous and not class-constant the right-hand side is $+\infty$ (Amjad & Geiger, 2020; Kolchinsky et al., 2019b), so no finite quantity bounds the deployed residual. This applies to FGIB as well (Proposition 4); what differs is structural, not a tightness ordering. In CEB the perturbation is *inside* the inference path and is driven by the prediction term, so removing it changes the model; in FGIB it lives on a side path that inference never reads and is controlled by side-path parameters alone. That is a cleaner separation of regularizer from model, and we claim nothing more for it.

C WHAT THE FROZEN REFERENCE DOES NOT BUY: FULL STATEMENT

(a) *The side head is a second trainable party — unless it is fixed.* The equivalence of Result 1 holds at each fixed A , and a trainable A , unused at inference — structurally the same position CEB’s backward encoder occupies on the other side of the KL. Since h is linearly separable by the jointly trained classifier, much of the anchor fit can be absorbed by A alone, and nothing in the objective resists A drifting toward a projector onto the K -dimensional between-class subspace. Full collapse is blocked (with A invertible, $\text{KL} = 0$ forces h itself to be class-constant), but the approach to the rank- K regime is not, and that is exactly where the conditioning of $A^\top A$ — which governs the size of the side-path perturbation in Result 1 — degrades without bound (Remark 7, Proposition 12(iii)). This channel is closed by the canonical configuration: a fixed orthogonal head has $\kappa(A^\top A) = 1$ by construction and anchors h directly (Result 3). The measurement of Section 4.5 quantifies how badly the *optional trainable* variant degrades on MNIST: $\kappa(A^\top A)$ rises to $10^{5.8}$ – $10^{6.8}$, i.e. severely ill-conditioned, with predictive accuracy unaffected — so the freedom buys nothing on MNIST; across five settings the fixed head pays ≈ 0.5 points per setting for the closure at the inherited scale, largely recovered by re-tuning the anchor scale (Table 8) — the accuracy-versus-certification trade-off made explicit. We therefore report the trainable head as the default of the released sweep but designate FGIB-H as the canonical, closed configuration.

(b) *The sweep needs a bounded classifier scale.* If c is free, then $(s, c) = (a, \infty)$ sends $\text{KL} \rightarrow 0$ and $\text{CE} \rightarrow 0$ simultaneously, so the infimum of L_β is 0 for every β and Result 4 has no global counterpart — the FGIB analogue of the dead knob, reached by growing the margin rather than by the linearity of the IB curve. Our classifier is an unconstrained `nn.Linear` trained by Adam without weight decay, so the proposition is conditional as implemented. A fixed-temperature cosine classifier satisfies the assumption by construction; weight decay only mitigates the divergence (Remark 12). One consequence matters for Section 4: since s^* drifts toward a as c grows, part of any observed β -effect under early stopping is plausibly an early-stopping effect, and we have not separated the two.

(c) *The certificate still does not bound the deployed residual.* P4 applies to FGIB unchanged (Proposition 4(b)). What FGIB buys is that the perturbation is side-path-only and its covariance is a known

Table 3: The anchor scale controls the fully compressed corner. Test score of FGIB at the strongest compression pressure in the grid ($\beta = 10$), as the anchor scale a varies; Δ is the gap from the deterministic backbone at the largest anchor ($a=16$). Units as in Table 1. At small a the anchors approach the single origin of a VIB prior and the model collapses to the trivial predictor (Remark 22); growing a recovers the backbone, and the 100-class task needs the largest a , as the $\ln(1 + (K - 1)e^{-ca})$ corner value predicts.

Task	Backbone	anchor scale a									Δ
		1	2	4	6	8	10	12	14	16	
<i>Classification (Acc %)</i>											
MNIST	MLP	98.1	98.0	98.2	98.3	98.3	98.3	98.3	98.3	98.3	-0.0
MNIST	CNN	98.1	99.0	99.2	99.2	99.3	99.2	99.3	99.2	99.3	+0.2
ImageNet-100	MLP	1.0	2.0	35.8	66.0	77.8	81.7	82.5	82.7	83.0	-0.2
ImageNet-100	CNN	81.0	82.5	83.7	84.2	84.2	84.4	84.7	84.3	84.6	+1.2
Cora	GCN	26.6	36.8	85.3	86.1	86.8	85.2	85.3	85.1	87.2	-0.5
IMDb	LSTM	86.0	86.1	85.6	85.7	85.4	85.7	86.3	85.7	85.3	+0.4
AG News	LSTM	92.7	92.8	92.7	92.7	92.6	92.7	92.7	92.7	92.7	-0.2
<i>Regression ($R^2 \times 100$)</i>											
Cal. Housing	MLP	76.5	77.7	79.1	79.2	79.2	79.3	79.1	79.2	78.6	+2.5
STS-B	LSTM	14.5	14.4	14.9	16.0	16.3	16.9	16.0	15.3	15.8	-13.2
ZINC-12k	GCN	60.2	58.4	58.2	57.5	58.4	59.2	59.5	59.2	58.9	-4.5
AgeDB	MLP	-0.1	-0.1	22.9	24.9	12.9	18.8	26.6	28.5	29.3	+7.5
AgeDB	CNN	17.4	18.4	24.6	30.9	33.9	35.3	36.7	35.9	35.6	-3.5

model parameter — a cleaner separation of regularizer from model, not a tighter or more valid bound (Remark 18).

(d) *The compressed endpoint is designed, not certified.* Caveat 2 of Kolchinsky et al. (2019a) identifies as uninteresting those variables that compress by mixing a constant map with a copy of Y ; FGIB’s compressed corner $\mu_y = av_y$ is a deterministic function of Y alone, i.e. a label code of exactly that kind. The orthonormal frame is in fact *opposed* to the semantic compression Caveat 2 holds up as interesting: it places every pair of classes at the same distance and can never express “merge these two classes, keep both far from that one”. What the anchors do is choose *one specific* label-code geometry — a neural-collapse-like configuration (Papayan et al., 2020) — instead of leaving it to the optimizer, and refuse the chance-level corner. Any claim that FGIB compresses *input* semantics at that endpoint is the claim Caveat 2 rules out (Remark 13).

(e) *The sweep is over a surrogate.* Result 4 concerns the $(\text{KL}_{\text{anchor}}, \text{CE})$ plane on the aligned family, not the information plane $(I(X; T), I(Y; T))$. On that family the posterior is class-constant, so $I(X; Z|Y) = 0$ for every s ; what varies along the sweep is anchor mismatch and classifier margin, not a proved compression of mutual information (Remark 11).

D ADDITIONAL TABLES

Table 4 repeats every cell of Table 1 with the standard deviation over the 5 runs, and records the hyperparameter that validation selected in each cell. Two patterns are worth noting. The prior bottlenecks concentrate on the smallest β in the grid (10^{-4} in 30 of the 60 non-FGIB cells, and $\beta \leq 10^{-3}$ in 45), which is the behavior a destructive knob induces: validation prefers the setting closest to no bottleneck at all. FGIB’s selected β instead spreads over 10^{-4} to 10^{-1} with 10^{-2} most common (6 of 12), and its selected anchor scale sits at the top of the grid ($a \geq 14$) on 4 settings and at the bottom ($a = 1$) on 3. Run-to-run variance is also informative: on the AgeDB/CNN setting the marginal-prior objectives have standard deviations of roughly 18 points of $R^2 \times 100$ (VIB ± 18.9 , NIB ± 18.9 , SVIB ± 18.7 , DVCCA ± 18.1) — some seeds collapse and others do not — while FGIB (± 1.4), the deterministic CNN (± 1.9) and CEB (± 2.8) are an order of magnitude tighter.

Table 5 reports the per-seed paired differences against the deterministic backbone at each objective’s validation-selected configuration, with bootstrap 95% confidence intervals over the 5 paired seeds. The per-setting means feed the Wilcoxon tests cited in Section 4.

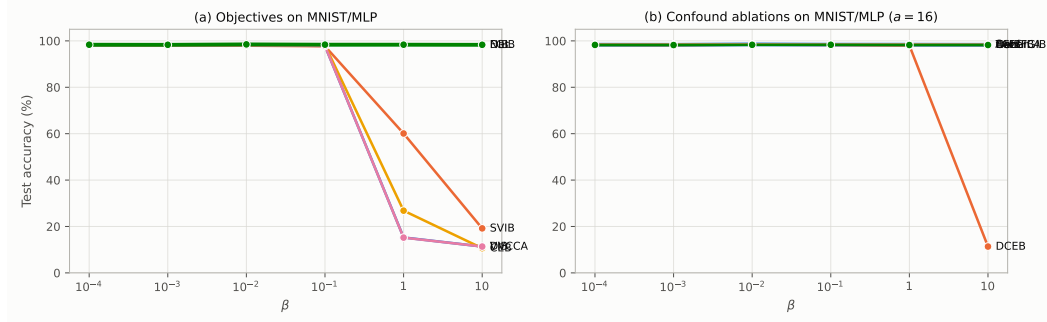


Figure 1: Test accuracy against β (logarithmic axis) on MNIST/MLP. (a) The seven objectives; (b) the confound ablations of Section 4.5 at $a = 16$. Lines are means over 5 seeds; every series is labeled at its right end.

Table 4: Standard deviations over the 5 runs for every cell of Table 1, with the validation-selected hyperparameters underneath each cell (β , and β, a for FGIB; -- for the deterministic backbone, which has neither). The last column is the equal-budget control: FGIB selected over the same 6-point β grid as every other bottleneck at a single globally fixed anchor scale $a = 16$ (a post-hoc sensitivity control), so its search budget matches the comparison (see Section 4). Units as in Table 1.

Task	Backbone	Det.	VIB	SVIB	NIB	CEB	DVCCA	FGIB	FGIB ($a=16$)
<i>Classification (Acc %)</i>									
MNIST	MLP	98.33 $\pm$ 0.12	98.28 $\pm$ 0.11	98.10 $\pm$ 0.06	98.26 $\pm$ 0.12	98.28 $\pm$ 0.09	98.24 $\pm$ 0.17	98.49 $\pm$ 0.12	98.49 $\pm$ 0.12
		--	0.001	0.001	0.001	0.0001	0.0001	0.01,16	0.01,16
MNIST	CNN	99.13 $\pm$ 0.05	99.28 $\pm$ 0.09	99.19 $\pm$ 0.07	99.19 $\pm$ 0.09	99.25 $\pm$ 0.10	99.28 $\pm$ 0.05	99.27 $\pm$ 0.04	99.27 $\pm$ 0.04
		--	0.01	0.0001	0.1	0.0001	0.01	0.001,16	0.001,16
ImageNet-100	MLP	83.12 $\pm$ 0.14	82.65 $\pm$ 0.32	83.02 $\pm$ 0.43	82.34 $\pm$ 0.27	82.56 $\pm$ 0.20	82.56 $\pm$ 0.43	83.96 $\pm$ 0.31	84.09 $\pm$ 0.18
		--	0.0001	0.0001	0.001	0.0001	0.0001	0.01,14	0.01,16
ImageNet-100	CNN	83.40 $\pm$ 0.13	82.94 $\pm$ 0.37	82.61 $\pm$ 0.53	82.72 $\pm$ 0.82	82.33 $\pm$ 0.38	83.09 $\pm$ 0.17	83.55 $\pm$ 0.42	83.63 $\pm$ 0.43
		--	0.0001	0.0001	0.01	0.0001	0.0001	0.001,12	0.001,16
Cora	GCN	87.68 $\pm$ 0.46	86.83 $\pm$ 0.28	77.56 $\pm$ 2.35	86.35 $\pm$ 0.68	86.27 $\pm$ 0.38	86.83 $\pm$ 0.38	87.68 $\pm$ 0.46	87.68 $\pm$ 0.46
		--	0.0001	0.001	0.0001	0.0001	0.0001	0.0001,1	0.0001,16
IMDb	LSTM	84.85 $\pm$ 0.68	85.43 $\pm$ 0.72	85.99 $\pm$ 0.30	86.11 $\pm$ 0.60	85.55 $\pm$ 0.73	85.69 $\pm$ 0.23	86.04 $\pm$ 0.40	85.54 $\pm$ 1.03
		--	0.0001	10	1	0.0001	0.0001	0.01,12	0.001,16
AG News	LSTM	92.94 $\pm$ 0.15	92.61 $\pm$ 0.21	92.70 $\pm$ 0.18	92.91 $\pm$ 0.22	92.66 $\pm$ 0.30	92.78 $\pm$ 0.15	92.71 $\pm$ 0.20	92.86 $\pm$ 0.11
		--	0.0001	0.0001	0.01	0.0001	0.0001	0.01,4	0.001,16
<i>Regression ($R^2 \times 100$)</i>									
Cal. Housing	MLP	76.18 $\pm$ 2.38	79.01 $\pm$ 0.49	78.97 $\pm$ 0.58	78.62 $\pm$ 0.35	78.94 $\pm$ 0.18	78.78 $\pm$ 0.48	79.85 $\pm$ 0.19	79.21 $\pm$ 0.07
		--	0.001	0.0001	0.01	0.0001	0.001	0.01,6	1,16
STS-B	LSTM	28.99 $\pm$ 1.13	27.19 $\pm$ 1.64	26.97 $\pm$ 2.02	24.17 $\pm$ 2.21	26.68 $\pm$ 1.27	27.05 $\pm$ 1.67	29.35 $\pm$ 0.96	29.35 $\pm$ 0.96
		--	0.01	0.001	0.01	0.01	0.001	0.001,16	0.001,16
ZINC-12k	GCN	63.48 $\pm$ 0.44	63.39 $\pm$ 0.33	63.25 $\pm$ 0.26	61.78 $\pm$ 0.99	62.57 $\pm$ 0.44	62.94 $\pm$ 0.26	63.42 $\pm$ 0.35	63.31 $\pm$ 0.35
		--	0.001	0.0001	10	0.0001	0.0001	0.1,1	0.0001,16
AgeDB	MLP	21.80 $\pm$ 4.09	31.73 $\pm$ 1.01	33.85 $\pm$ 2.86	27.10 $\pm$ 2.35	33.17 $\pm$ 2.51	31.80 $\pm$ 2.28	32.71 $\pm$ 1.57	29.43 $\pm$ 8.14
		--	0.001	0.0001	10	0.01	0.001	0.1,1	1,16
AgeDB	CNN	39.14 $\pm$ 1.94	33.42 $\pm$ 18.94	33.23 $\pm$ 18.70	33.37 $\pm$ 18.89	38.82 $\pm$ 2.75	32.05 $\pm$ 18.11	39.56 $\pm$ 1.39	35.93 $\pm$ 0.94
		--	0.001	0.0001	1	0.001	0.001	0.01,10	1,16

Table 6 reports the sensitivity of the equal-budget FGIB rank to the fixed anchor scale (Section 4). The remaining tables and figures of Section 4.5 follow.

E CONFOUND ABLATIONS: FULL DETAIL

What FGIB’s robustness is made of. FGIB differs from the comparison in two coupled ways: its classifier consumes the deterministic h at training and evaluation (P3, P4), and its prior is frozen (P1, P2). Table 10 separates the contributions. VIB and CEB fall to chance at $\beta = 1$ (15.3% and 26.8%); DCEB — the deterministic path of FGIB with CEB’s trainable backward encoder as its prior — holds at ≈ 98 through $\beta = 1$ and falls to 11.4% only at $\beta = 10$. Corollary 1 explains *why* DCEB

Table 5: Paired per-seed differences against the deterministic backbone, at each objective’s validation-selected configuration (Table 1), with bootstrap 95% confidence intervals over the 5 paired seeds (10,000 resamples). Units are percentage points (accuracy for classification, $R^2 \times 100$ for regression); positive means the objective beats the backbone at that setting. The last column is the equal-budget FGIB control of Table 4. Wilcoxon signed-rank tests over the 12 setting-level paired means against zero are reported in the text as an *exploratory* summary: the 12 settings mix classification and regression metrics, are not independent (shared backbones and protocols), and the 5 seeds describe training noise, not split uncertainty (a single fixed split per dataset); the classification-only (7 settings) and regression-only (5 settings) tests are reported alongside.

Task	Backbone	VIB	SVIB	NIB	CEB	
<i>Classification (Acc %)</i>						
MNIST	MLP	-0.05 [-0.15,+0.05]	-0.23 [-0.31,-0.15]	-0.06 [-0.16,+0.03]	-0.05 [-0.20,+0.05]	-0.05
MNIST	CNN	+0.14 [+0.06,+0.22]	+0.06 [+0.00,+0.12]	+0.06 [-0.04,+0.14]	+0.12 [+0.07,+0.17]	+0.12
ImageNet-100	MLP	-0.48 [-0.70,-0.25]	-0.10 [-0.44,+0.31]	-0.78 [-0.92,-0.64]	-0.56 [-0.67,-0.44]	-0.56
ImageNet-100	CNN	-0.47 [-0.79,-0.13]	-0.80 [-1.19,-0.35]	-0.69 [-1.29,-0.21]	-1.07 [-1.29,-0.85]	-1.07
Cora	GCN	-0.85 [-1.33,-0.41]	-10.11 [-11.85,-8.12]	-1.33 [-2.10,-0.56]	-1.40 [-1.88,-0.81]	-1.40
IMDb	LSTM	+0.58 [-0.31,+1.41]	+1.15 [+0.62,+1.69]	+1.26 [+0.76,+1.63]	+0.70 [-0.11,+1.75]	+0.70
AG News	LSTM	-0.33 [-0.61,-0.14]	-0.24 [-0.47,-0.02]	-0.03 [-0.27,+0.20]	-0.28 [-0.62,+0.04]	-0.28
<i>Regression ($R^2 \times 100$)</i>						
Cal. Housing	MLP	+2.83 [+0.87,+4.93]	+2.79 [+0.89,+4.83]	+2.44 [+0.72,+4.51]	+2.76 [+0.93,+4.75]	+2.76
STS-B	LSTM	-1.81 [-3.09,-0.41]	-2.02 [-3.75,+0.00]	-4.83 [-6.40,-3.32]	-2.31 [-3.63,-1.11]	-2.31
ZINC-12k	GCN	-0.10 [-0.51,+0.47]	-0.23 [-0.47,-0.00]	-1.70 [-2.76,-0.80]	-0.92 [-1.30,-0.42]	-0.92
AgeDB	MLP	+9.93 [+6.42,+13.43]	+12.05 [+10.05,+13.64]	+5.30 [+3.09,+7.50]	+11.37 [+6.91,+15.93]	+11.37
AgeDB	CNN	-5.72 [-22.83,+3.64]	-5.91 [-22.78,+3.19]	-5.78 [-22.75,+3.61]	-0.32 [-2.82,+2.18]	-0.32

Table 6: Sensitivity of the equal-budget FGIB rank to the fixed anchor scale. Each column reports FGIB selected over the same 6-point β grid as every other bottleneck at a *single globally fixed* anchor scale a , so its search budget matches the comparison exactly (the $a=16$ column is the post-hoc control of Table 4). Mean rank of 7 entries (deterministic backbone + 6 bottlenecks) over all 12 settings, over the 7 classification settings, and over the 5 regression settings; “best” counts best-or-tied settings out of 12. The main-table number (1.75, full 6×9 grid) is a ceiling only over the full grid, not over fixed a ; the range of fixed- a ranks quantifies how much of FGIB’s edge is anchor search budget (Section 4).

anchor scale a	1	2	4	6	8	10	12	14	16
mean rank (12 settings)	3.17	2.75	3.17	2.92	2.75	2.25	2.33	2.00	2.33
mean rank (classification, 7)	3.29	2.43	2.86	2.29	2.43	2.14	1.86	1.57	2.14
mean rank (regression, 5)	3.00	3.20	3.60	3.80	3.20	2.40	3.00	2.60	2.60
best or tied-best (of 12)	3	2	4	4	4	6	5	7	6

is the odd one out: its trainable prior *variance* has a negative gradient in the mean gap, so the race of Channel P3 re-enters through the prior and amplifies the mean-matching force — the only FGIB-family variant that escapes the exact center-loss family is the only one that collapses. By contrast, every variant that *starts* from the orthonormal anchor table stays at ≈ 98.3 across the entire grid: *TAFGIB* (trainable anchor means), *CentFGIB* (KL replaced by the plain center loss $\frac{1}{2}\|\mu - av_y\|^2$), FGIB with A frozen after random initialization, and *FGIB-H* with $A=I$. The equivalence is not a numerical coincidence: by Result 2 the zero-initialized variance head receives zero gradient at every step, so FGIB’s KL is the center loss exactly, and CentFGIB, TAFGIB and FGIB-H are the same objective up to the trainability of the anchor means. The trainable anchors of TAFGIB do move (final column norms shrink from 16 to 5.7–10, pairwise correlations reach 0.78 at the smallest β), yet the accuracy curve is untouched — the protective property is the anchor-table *initialization* geometry at scale a , not the orthonormality of the trained table. Two conclusions follow. First, the load-bearing ingredient on MNIST is that initialization geometry — orthonormal per-class directions at scale a — not the frozenness of its means, and not any Gaussian variance machinery, which is provably inert here (Result 2). Second, freezing the means and keeping the KL is what the *theory*

Table 7: CentFGIB across all 12 settings: does the geometry alone — the same orthonormal anchors as FGIB but with the KL replaced by a plain $\frac{1}{2}\|\mu - av_y\|^2$ center loss — reproduce FGIB’s behavior beyond MNIST? Validation-selected configuration at $a = 16$ (the same 6-point β grid and budget as the FGIB equal-budget control of Table 4), 5 seeds, protocol identical to the main sweep. Units as in Table 1.

Task	Backbone	Det.	FGIB ($a=16$)	CentFGIB ($a=16$)
MNIST	MLP	98.33	98.49	98.49
MNIST	CNN	99.13	99.27	99.26
ImageNet-100	MLP	83.12	84.09	84.09
ImageNet-100	CNN	83.40	83.63	83.63
Cora	GCN	87.68	87.68	87.68
IMDb	LSTM	84.85	85.54	85.54
AG News	LSTM	92.94	92.86	92.86
Cal. Housing	MLP	76.18	79.21	79.21
STS-B	LSTM	28.99	29.35	29.35
ZINC-12k	GCN	63.48	63.31	63.32
AgeDB	MLP	21.80	29.43	29.43
AgeDB	CNN	39.14	35.93	35.50

Table 8: The fixed-orthogonal side head across settings. FGIB, CentFGIB and FGIB-H differ only in the side-path regularizer — trainable side head + KL, trainable side head + plain center loss, and fixed side head $A=I$ (equivalently the anchor matching applied directly to the deployed representation h , Result 3) — each selected over the same 6-point β budget at a single fixed anchor scale, on validation. $a=16$ is the scale inherited from the z -space sweep; $a=4$ is a re-tuned scale for h -space (the fixed head’s adaptive rescaling is gone, so its optimal scale shifts). Units as in Table 1. At the inherited scale the fixed head trails by ≈ 0.5 points per setting; at the re-tuned scale it recovers most of the gap and ties or beats the backbone on every row — the price of the $\kappa=1$ closure is small, and both configurations ship in the released code.

Task	Backbone	Det.	FGIB ($a=16$)	CentFGIB ($a=16$)	FGIB-H ($A=I, a=16$)	FGIB-H ($A=I, a=4$)
MNIST	MLP	98.33	98.49	98.49	98.26	—
MNIST	CNN	99.13	99.27	99.26	99.24	99.29
ImageNet-100	MLP	83.12	84.09	84.09	83.52	83.45
Cal. Housing	MLP	76.18	79.21	79.21	78.08	78.78
Cora	GCN	87.68	87.68	87.68	86.68	87.68
Mean rank (of 5)		3.80	1.20	1.40	4.00	2.25
Best or tied-best (of 5)		1	4	4	0	2

buys — the persistent force of Remark 17, the convexified sweep of Proposition 8, the residual-information reading of Result 1 — so FGIB retains an exact variational upper-bound interpretation for the auxiliary noisy code; its predictive robustness and endpoint behavior outside the analyzed family remain empirical. Constraining the side head changes nothing in the β -profile (98.2–98.4 across the grid for frozen A and $A=I$): the A -channel degrades $\kappa(A^\top A)$ but does not carry the robustness.

The variance race does not materialize. We ran the falsification test that Proposition 3 invites, logging the mean posterior log-variance, the clamp fraction (e^{-10}), and (for CEB) the prior log-variance at every epoch of every run, aggregated over the *entire* validation set, not a single batch. At the final epoch, averaged over the 5 seeds, CEB’s posterior log-variance sits at $+0.37, +0.27, +0.13, +0.08, -0.05, -0.01$ (nats) for $\beta = 10^{-4}, \dots, 10$ — a mild monotone drift with β , nowhere near the clamp — and the clamp fraction is 0.0 at *every* epoch of every run; CEB’s prior tracks its posterior, and VIB’s posterior stays within 0.37 of zero (Figure 2(b)). To probe whether the race is an early-stopping artifact, we retrained CEB at $\beta \in \{1, 10\}$ for 400 epochs with early stopping disabled. The answer is nuanced and we report it as measured: the variance *does* drift, monotonically and in exactly the direction Proposition 3(i)–(ii) predicts — at $\beta = 1$ the posterior log-variance ends at -2.27 nats (-1.20 at $\beta = 10$; the prior tracks the posterior), passing

Table 9: AgeDB under a subject-disjoint split. Identities are grouped so that no subject appears on both sides of any split boundary (60/20/20, seed 42); this removes the identity leakage of the image-level split used elsewhere in the paper, at the cost of comparability with other AgeDB literature. Validation-selected β at a fixed $a = 16$ (6-point grid), 5 seeds, CNN backbone, $R^2 \times 100$.

AgeDB (subject-disjoint)	Det.	FGIB ($a=16$)	CentFGIB ($a=16$)	FGIB-H ($A=I$, $a=16$)
$R^2 \times 100$ (val-sel. β)	21.37	35.10	35.10	33.71
selected β	—	0.001	0.001	1

Table 10: Confound ablations and fixed-scale checks on MNIST/MLP. Test accuracy (%) across the β grid for each variant (fixed $a = 16$ where an anchor scale applies), plus the validation-selected configuration. *DCEB* keeps the deterministic classifier path of FGIB but trains the conditional prior (CEB’s backward encoder); *TAFGIB* trains the anchor means instead of freezing them; *CentFGIB* replaces the KL with a direct MSE-to-anchor center loss; *frozen A* and $A=I$ constrain the side head; *cosine* rows use the fixed-temperature cosine classifier of Remark 12(c). Protocol identical to the main experiments (100 epochs, patience 10, 5 seeds).

Variant	10^{-4}	10^{-3}	10^{-2}	10^{-1}	1	10	val-sel.
Det.	—	—	—	—	—	—	98.33
VIB	98.18	98.28	98.18	97.83	15.27	11.35	98.28
CEB	98.28	98.30	98.43	98.16	26.82	10.64	98.28
FGIB ($a=16$)	98.35	98.31	98.49	98.37	98.36	98.31	98.49
DCEB (trainable prior)	98.26	98.26	98.38	98.24	98.04	11.35	98.26
TAFGIB (trainable anchors)	98.22	98.31	98.42	98.42	98.41	98.33	98.42
CentFGIB (MSE-to-anchor)	98.35	98.31	98.49	98.37	98.36	98.31	98.49
FGIB frozen A	98.32	98.22	98.37	98.35	98.26	98.27	98.32
FGIB $A=I$	98.26	98.20	98.30	98.32	98.21	98.21	98.26
FGIB cosine	97.94	98.21	98.32	98.30	98.34	98.30	98.34
VIB cosine	97.93	98.10	98.12	96.56	51.85	11.42	98.10
CEB cosine	97.99	98.35	98.25	41.63	10.79	10.31	98.35
ETF (frozen classifier)	—	—	—	—	—	—	98.32

—0.92 at epoch 100 — but the clamp fraction is 0.0 throughout, and -2.3 nats ($\sigma^2 \approx 0.10$) is still far from the e^{-10} clamp the two-stage account predicts as the terminal state (Figure 3). The race is therefore *real but slow*: at the protocol horizon it is invisible, at four times the horizon its direction is unmistakable, and its convergence time-scale is beyond 400 epochs at these hyperparameters. That is precisely the conditional status Remark 4 assigns to Proposition 3(iii): a direction-of-force statement whose terminal state our longest run has not reached. For FGIB the question is settled analytically: by Result 2 the posterior variance is pinned at τ^2 at every step — the flat zero line of Figure 3 is the fixed point itself, held for 400 epochs — and Corollary 1 identifies the trainable prior variance as the only re-entry point for the race.

The side-head conditioning channel. Section 3.3(a) identifies the *trainable* side head A as the channel the optional variant leaves open, and predicts that the approach to a rank- K projector would show up as diverging $\kappa(A^\top A)$. Figure 2(a) plots $\log_{10} \kappa(A^\top A)$ over training, at $a = 16$, for every β : from its initialization near $10^{3.9}$ the condition number rises to $10^{5.8}$ – $10^{6.8}$ at the final epoch, with transient peaks to $10^{8.3}$ — $A^\top A$ becomes *severely ill-conditioned* ($\approx 1000\times$ degradation, no monotone dependence on β). Predictive accuracy nevertheless remains stable, so the conclusion is narrow: the ill-conditioning does not hurt MNIST predictions in this experiment, but it means the pulled-back reference of Result 1 is not close to a well-conditioned frame, and the KL’s correspondence with h is not guaranteed by anything we measured for the trainable variant. The canonical configuration needs no such caveat: $A = I$ has $\kappa = 1$ at every step and anchors h directly (Result 3). The $A=I$ and frozen- A rows of Table 10 show what the degradation costs in accuracy (little on MNIST), and Table 8 quantifies the fixed head’s price across five settings: ≈ 0.5 points per setting at the inherited anchor scale, largely recovered by re-tuning the scale — a real but bounded margin.

Method	Fixed geometry	Zero-penalty set
VIB (Alemi et al., 2017)	—	collapse (origin)
CEB (Fischer, 2020)	—	underdetermined family
center loss (Wen et al., 2016)	learned centers	no bottleneck reading
HPN (Mettes et al., 2019)	hypersphere (a priori)	—
fixed classifier (Hoffer et al., 2018; Zhu et al., 2021; Pernici et al., 2019)	simplex/ETF rows	—
FGIB (ours)	orthonormal anchors (a priori)	prescribed class codes

Table 11: FGIB against the closest fixed-geometry and prototype regularizers. “Zero-penalty set” is the set of representations where the regularizer vanishes. “Variational bound” asks whether the regularizer carries a KL/information interpretation. “Regularizer on deployed repr.” asks whether the regularized object is the representation inference consumes.

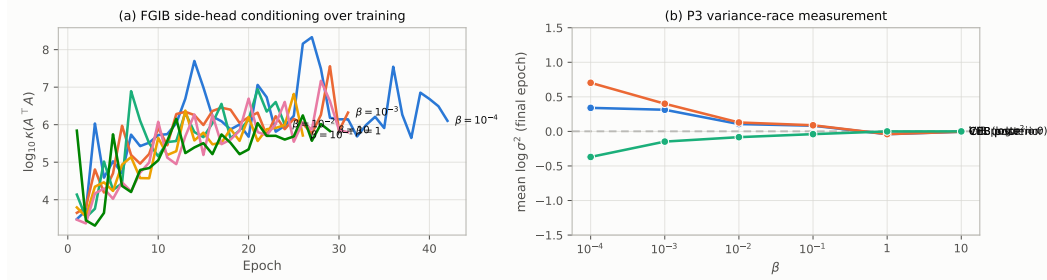


Figure 2: Diagnostics on MNIST/MLP. (a) $\log_{10} \kappa(A^T A)$ of the FGIB side head over training, one seed per β . (b) The P3 variance-race measurement: mean log-variance at the final epoch against β for CEB’s posterior and prior and VIB’s posterior; the dashed line is the initialization $\log \sigma^2 = 0$.

Fixed classifier scale. Proposition 8 requires a fixed classifier scale c , which the default `nn.Linear` does not enforce. The cosine rows of Table 10 enforce it by construction: FGIB’s sweep is unchanged (97.9–98.3 across all six β values), while VIB and CEB are not rescued (51.9% at $\beta = 1$ and 41.6% at $\beta = 0.1$ respectively), exactly as Remark 12 predicts — fixed c does not repair the trainable-prior corners; the anchor geometry, not the free classifier scale, keeps FGIB’s corner non-degenerate (Appendix D, cosine rows).

Does the geometry alone generalize? Result 2 proves the KL machinery is *never* load-bearing in this architecture, on any dataset. Table 7 asks the cross-dataset version: CentFGIB run on all 12 settings under the equal-budget protocol ($a = 16$, 6-point β grid, 5 seeds). The geometry alone generalizes: at the validation-selected configuration, CentFGIB matches FGIB($a=16$) to within 0.01 points on 10 of 12 settings and to within 0.43 on the remaining two, including all four regression settings with RFF anchors; in a nine-entry ranking the FGIB family occupies the top three slots. The empirical engine of the method is the fixed anchor geometry itself; the KL contributes the bound reading and the sweep theory (Appendix D).

Does the side head need to exist at all? The canonical configuration FGIB-H ($A=I$) removes the side head from the theory’s concerns entirely: the anchor matching acts directly on the deployed representation (Result 3). Table 8 measures the price across five settings — MNIST/MLP and /CNN, ImageNet-100 features, California Housing, Cora. At $a = 16$, the scale inherited from the z -space sweep, the fixed head trails the trainable one by ≈ 0.5 points per setting (mean rank 4.00 of 5; FGIB 1.20, CentFGIB 1.40) — the trainable head is doing real adaptive work, and part of that work is rescaling the effective geometry. Re-tuning the anchor scale for h -space recovers most of the gap: at $a = 4$, FGIB-H takes mean rank 2.25 and ties or beats the backbone on every row (MNIST/CNN 99.29, best of the table; Cora 87.68, tied-best), so the price of the $\kappa = 1$ closure is small and quantifiable — an accuracy-versus-certification choice whose two sides both ship in the released code. We also test the other fixed geometry the class invites: a backbone with a *frozen ETF classifier* (equiangular tight frame head, no regularizer) scores 98.32% on MNIST/MLP against the backbone’s 98.33% — a fixed classifier alone buys neither margin nor a knob, so the anchor regularization on the representation is the load-bearing ingredient (Table 10, ETF row).

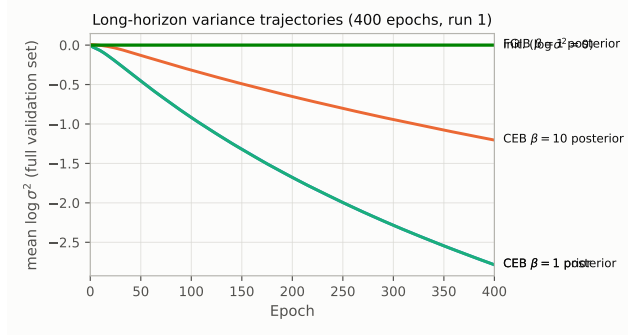


Figure 3: Long-horizon variance trajectories on MNIST/MLP (400 epochs, early stopping disabled, one seed, full validation set per epoch). CEB’s posterior and prior log-variances at $\beta=1$ and $\beta=10$ stay near the initialization $\log \sigma^2 = 0$; FGIB’s posterior is the flat zero line — the fixed point of Result 2, held exactly for the whole run.

AgeDB without identity leakage. The image-level AgeDB split of the main sweep leaks identity across the boundary. To check that the leak inflates magnitudes rather than creates the pattern, we reran AgeDB/CNN under a subject-disjoint split: the ordering is preserved — the backbone scores 21.37 against 35.10 for FGIB and CentFGIB and 33.71 for FGIB-H (all $R^2 \times 100$, validation-selected β) — so the leakage does not manufacture the regression claim (Table 9, Appendix D).

Anchor-scale sensitivity of the equal-budget rank. The $a=16$ equal-budget control is a post-hoc choice; Table 6 reports the fixed-anchor rank for every a in the grid: the mean rank over the 12 settings ranges from 2.00 ($a=14$, best or tied-best on 7) to 3.17 ($a=1$), with $a=16$ at 2.33 — the edge is not an artifact of the chosen anchor, and any $a \geq 8$ beats every bottleneck’s rank. The remaining tables and diagnostic figures of this section are collected in Appendix D.

F WHAT WE HAVE NOT MEASURED: FULL STATEMENT

(i) *Separating the β -effect from the early-stopping effect.* Remark 12(b) predicts that the β -dependence of the equilibrium decays with training length; our runs use patience 10, so training well past saturation and comparing converged KL across β would test it. The cosine-classifier runs of Section 4.5 enforce the fixed- c assumption of Proposition 8 but do not remove early stopping, and the 400-epoch control of that section covers the variance trajectory, not the sweep itself. (ii) *Equal budget for the other methods.* The $a = 16$ column of Table 4 gives FGIB the same 6-point budget as the comparison; Table 6 shows the rank is stable across fixed anchor scales, not only at $a = 16$. The converse — giving VIB/CEB/DVCCA an extra temperature or variance hyperparameter dimension — is not run, so the comparison’s rank ordering is only shown to survive budget restriction for FGIB, not budget expansion for the others. (iii) *The remaining protocol asks.* All results come from one fixed 60/20/20 split per dataset (seed 42), so split-sensitivity is unmeasured — the subject-disjoint AgeDB control measures leakage sensitivity for one dataset, not split noise in general. Beyond the theory’s own asks, the robustness properties that motivate conditional bottlenecks in the first place — adversarial robustness, OoD detection, calibration, and resistance to memorizing random labels (Fischer, 2020; Fischer & Alemi, 2020) — are not evaluated here; the label-free anchor-distance score $\min_y \text{KL}(q(z|x) || \mathcal{N}(av_y, \tau^2 I))$ that FGIB’s geometry makes available at inference (Remark 6) is a hypothesis, not a result.

G DATASETS AND PREPROCESSING

All splits use seed 42. Except where noted, each dataset is divided 60/20/20 into train/validation/test; classification splits are stratified, regression splits are not. Targets for regression are min-max scaled to $[0, 1]$ with a scaler fitted on the training split only, and MAE is reported in original units via the inverse transform.

MNIST (LeCun et al., 1998): 70,000 28×28 grayscale digits; the 60,000-image training partition is split into 50,000 train / 10,000 validation, the official 10,000-image test set is untouched. Inputs are normalized to $[0, 1]$.

ImageNet-100 (Deng et al., 2009): frozen ResNet-50 (He et al., 2016) layer-3 features. Two variants are used: global average pooling (1024-d, the MLP backbone) and 4×4 spatial pooling ($1024 \times 4 \times 4$, the CNN backbone). Features are split 60/20/20 (stratified) and standardized with a StandardScaler fitted on the training split.

Cora (Sen et al., 2008): the standard transductive node-classification graph (2708 nodes, 1433-d features, 7 classes) with a fixed random split (seed 42) into train/validation/test node masks; the adjacency matrix is symmetrized, self-loops are added, and the normalized Laplacian-style $\tilde{D}^{-1/2}(\tilde{A})\tilde{D}^{-1/2}$ is used. Training is full-graph (transductive), following standard GCN practice (Kipf & Welling, 2017); this is not the fixed Planetoid split, so absolute numbers are not directly comparable with published Planetoid numbers — all comparisons in this paper are internal to the same split.

IMDb (Maas et al., 2011): the official 50,000 labeled reviews (25k train + 25k test) are merged and re-divided 60/20/20 (stratified) into 30k/10k/10k. HTML tags are stripped, tokens are alphabetic words from a training-split vocabulary (approx. 72k words at length cap 250), and sequences are truncated or padded to 250 tokens. Official-split literature numbers are therefore not directly comparable; again, all comparisons are internal.

AG News (Zhang et al., 2015): frozen bert-base-uncased (Devlin et al., 2019) token features. We use the second-to-last hidden layer (layer 10) of each token, PCA-reduced to 50 dimensions (the PCA fit on training-split tokens only), sequences capped at 64 tokens; 60/20/20 stratified split over 127,600 documents.

California Housing (Pace & Barry, 1997): 20,640 samples, 8 features, 60/20/20 split (regression, unstratified); features standardized on the training split, target min-max scaled.

STS-B (Cer et al., 2017; Wang et al., 2019): the official GLUE train and dev partitions (7,249 labeled sentence pairs, similarity 0–5) are merged and re-divided 60/20/20 into 4,349/1,450/1,450 (the GLUE test partition carries no labels and is unused). Each sentence is stripped of HTML, tokenized alphabetically, and truncated to 124 tokens; the pair is joined by a [SEP] token into a single sequence of length ≤ 250 . The vocabulary is built from the training split (approx. 10k words). Embeddings are GloVe-6B-100d (Pennington et al., 2014), initialized and frozen; the LSTM has a single 256-d hidden layer and max-pooling readout.

ZINC-12k (Dwivedi et al., 2023): the official benchmarking-gnns split (10,000 train / 1,000 validation / 1,000 test). Node features are 28-d one-hot atom types (bond types are unused, matching the benchmarking GCN baseline); the adjacency is symmetrized, self-loops added, and degree-normalized as for Cora; batching uses block-diagonal adjacency (batch size 128). The target is the penalized logP, min-max scaled on the training split; MAE is reported in original units (our GCN baseline reproduces the literature value to within the expected margin).

AgeDB (Moschoglou et al., 2017): the official full release, 16,488 images of 567 subjects aged 0–96, resized to RGB 64×64 and normalized with ImageNet statistics; target age min-max scaled on the training split. **Split caveat:** the default split is image-level 60/20/20, under which 562 of 567 subjects have images on both sides of the train/test boundary — identity-specific age cues can leak, and the AgeDB numbers should be read with that in mind. A subject-disjoint split (grouping all images of a subject into one partition, then splitting subjects 60/20/20 with seed 42) is implemented in the released code and *is run* as a control in this paper: under it the objective ordering is preserved (Table 9), so the leak inflates AgeDB’s magnitudes but does not create the pattern.